Shimura varieties

Xinwen Zhu

Lecture 1

k field

19 centuries, $f \in k[x]$ $\quad E|k$ $\quad$ Galois group $\quad G_f$ of f

$$G_f = \mathrm{Gal}(E|k) \curvearrowright E = k(\alpha_1, \dots, \alpha_n)$$

$$\mathrm{Gal}(\bar{k}|k) = \varprojlim_{E \subset \bar{k}} \mathrm{Gal}(E|k) \qquad C(\text{root}) = H^0((\mathrm{Spec}\, E)_{\bar{k}})$$

20 centuries, system of (homogeneous) poly eqns (smooth proj.) $X \subset \mathbb{P}^n_k$

Grothendieck's conj $\quad G_X$ motivic Galois group

pro-alg gp $/\mathbb{Q}$

$$1 \to G_X^\circ \to G_X \to \mathrm{Gal}(\bar{k}|k) \to 1$$

Although we don't know what G_X are, we could try to understand the rep'n of G_X. rep'n should come from cohomology of X.

$$X/k \rightsquigarrow H_M(X/k) \begin{cases} H_{dR}(X/k) \\ \updownarrow \\ H_B(\sigma X) \\ \updownarrow \\ H_{\acute{e}t}(X_{\bar{k}}) \end{cases} \text{vector space w/ additional str.}$$

$k \hookrightarrow \mathbb{C}$

Comparison theorem

$$H_{dR}(X/k) \underset{k,\sigma}{\otimes} \mathbb{C} \xrightarrow{\sim} H_B(\sigma X, \mathbb{Q}) \otimes \mathbb{C}$$

$$\Big\}$$

rat'l Hodge structure h

which can be regarded as rep'ns of Mumford-Tate gp $G_{MT}(h)$

What are Shimura varieties?

They (should be) are moduli spaces of varieties (motives) X s.t. $G_X^\circ \subset$ fixed (reductive) alg. gp G.

In reality, they are moduli spaces of (rat'l) Hodge str. h

s.t. $G_{MT}(h) \subset G$.

Ex. (1) $\left\{ \begin{array}{l} X: \\ y^2 = x^3 + ax + b \end{array} \right\}$ elliptic curve

$\mathbb{P}^1$ $\quad G_X^\circ = \begin{cases} GL_2 \\ \mathrm{Res}_{E|\mathbb{Q}} \mathbb{G}_m \end{cases}$ $\quad$ E quadratic imaginary field

moduli of elliptic curves $\quad SL_2(\mathbb{Z}) \backslash \mathcal{H}$

(2) $\{ y^2 = x(x-1)(x-\lambda) \}$

$\downarrow$

$\mathbb{A}^1 - \{0,1\}$

$\Gamma(2) \backslash \mathcal{H}$

(3) $y^3 = x(x-1)(x-\lambda_1)(x-\lambda_2)$

$\downarrow$

$\mathbb{A}^2 - \{\cdots\}$

$G_X = U(2,1)$

Picard modular surface .....

Hodge structure.

Fix $\Lambda (\simeq \mathbb{Z}, \mathbb{Z}[\frac{1}{p}], \mathbb{Z}_{(p)}, \mathbb{Q}, \mathbb{R}) \subset \mathbb{R} \hookrightarrow \mathbb{C}$

Dedekind domain

<u>Def</u> A pure Λ-HS of wt m is a finite free Λ-module V together w/ a decreasing filtration $F^\bullet(V_\mathbb{C})$ of $\mathbb{C}$-v.s. $F^i(V_\mathbb{C}) \supset F^{i+1}(V_\mathbb{C}) \supset \cdots$

s.t. (1) $\bigcap F^i V_\mathbb{C} = 0$

$F^i V_\mathbb{C} = V_\mathbb{C}$ for $i \ll 0$

(2) $\forall p \in \mathbb{Z}$, $F^p V_\mathbb{C} \oplus \overline{F^{m-p+1} V_\mathbb{C}} = V_\mathbb{C}$

$\ulcorner$ $\overline{(\cdot)} : V_\mathbb{C} \xrightarrow{\sim} V_\mathbb{C}$

$W \subset V_\mathbb{C} = \mathbb{C} \otimes_\Lambda V$

$\overline{W} = \{ \overline{v} : v \in W \}$, $\overline{W} \subset V_\mathbb{C}$ a $\mathbb{C}$-v.s. $\lrcorner$

$$\{ (V, F^\bullet) \text{ pure of wt } m \} \xrightarrow{\sim} \left\{ \left(V, V_\mathbb{C} = \bigoplus V_\mathbb{C}^{p,q} \text{ decomp. of } \mathbb{C}\text{-v.s.}\right) \text{ s.t. } \begin{cases} V_\mathbb{C}^{q,p} = \overline{V_\mathbb{C}^{p,q}} \\ V_\mathbb{C}^{p,q} = 0 \text{ unless } p+q=m \end{cases} \right\}$$

$$(V, F^\bullet) \longmapsto \left(V, V^{p,q} := F^p V_\mathbb{C} \cap \overline{F^{m-p} V_\mathbb{C}}\right)$$

$$F^p V_\mathbb{C} = \bigoplus_{p' \geq p} V^{p',q} \longleftarrow\!\shortmid V_\mathbb{C} = \bigoplus V^{p,q}$$

Ex. Let $X/\mathbb{C}$ be a smooth proj. var. of dim n

$$0 \to \mathcal{O}_X \xrightarrow{d} \Omega^1_X \xrightarrow{d} \Omega^2_X \to \cdots \to \Omega^n_X \to 0$$

$$H^i_{dR}(X/\mathbb{C}) := \mathbb{H}^i((\Omega^\cdot_X, d))$$

$$\wr\downarrow$$

$$H^i_B(X^{an}, \mathbb{C}) \supset H^i_B(X^{an}, \Lambda)/\text{torsion}$$

$$\| $$

$$H^i_B(X^{an}, \Lambda) \otimes_\Lambda \mathbb{C}$$

$$F^{p+1}((\Omega_X, d)) = \left[0 \cdots \to 0 \to \Omega^{p+1}_X \to \Omega^{p+2}_X \to \cdots \to \Omega^n_X \to 0 \right]$$

$$\downarrow \quad \|$$

$$\Omega^p_X \to \Omega^{p+1}_X \to \cdots$$

$$\mathrm{Gr}^p_F(\Omega_X, d) = \Omega^p_X[-p]$$

$$\mathbb{H}^m(F^p(\Omega_X, d)) \longrightarrow \mathbb{H}^m((\Omega_X, d))$$

$\rightsquigarrow$ Hodge-to-de Rham spectral sequence

$$E_1^{p,q} = H^q(X, \Omega^p) \Rightarrow \mathrm{Gr}_F H^m_{dR}(X/\mathbb{C})$$

Thm (degeneration of H-to-dR ss)

concretely

$$\begin{tikzcd}
 & & H^{m-p}(\Omega^p) \arrow[d, equal] & \\
\to H^m(F^p) \arrow[r, hook] & H^m(F^{p-1}) \arrow[r, two heads] \arrow[d] & H^m(\Omega^p[-p]) \arrow[r] & \\
 & H^m_{dR}(X/\mathbb{C}) & &
\end{tikzcd}$$

(Hodge theory)

Thm $(H^m_B(X, \Lambda), F^\cdot)$ is a pure Hodge str. of wt m

<u>Sub-Ex</u> X curve of genus g

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{O} \arrow[r] & \Omega^1_X \arrow[r] & 0 & F^1 \\ 0 \arrow[r] & \mathcal{O}_X \arrow[r] \arrow[u] & \Omega^1_X \arrow[r] \arrow[u] & 0 & F^0 \\ 0 \arrow[r] & \mathcal{O}_X \arrow[r] \arrow[u] & \mathcal{O} \arrow[r] \arrow[u] & 0 & Gr^0_F \end{tikzcd}$$

$$\begin{array}{c} H^1(X,\Lambda) \\ \downarrow \\ H^1(X,\mathbb{C}) \\ \wr\wr \\ H^0(X,\Omega_X) \hookrightarrow H^1_{dR}(X,\mathbb{C}) \twoheadrightarrow H^1(X,\mathcal{O}_X) \end{array}$$

$$\overline{H^0(X,\Omega_X)} \cap H^0(X,\Omega_X) = 0$$

$g=1$ $\quad X = \{y^2 = x^3 + ax + b\}$

$\omega = \dfrac{dy}{x}$ $\quad \int_{\gamma_1} \omega \quad \int_{\gamma_2} \omega$ can't be both real numbers

<u>Ex</u> Let $X = A$ abelian var. $/\mathbb{C}$, $X^{an} = \mathbb{C}^g / L$

Cat HS_Λ of Λ-HS

$\Lambda \subset R$ some ring R.

<u>Def</u> $\mathrm{Fil}_{R/\Lambda}$ obj. : finite free Λ-mod V, decreasing filtration $F^\bullet V_R$

$$\cdots \supset F^p V_R \supset F^{p+1} V_R \supset \cdots$$

s.t. $\begin{cases} F^i = 0, & i \gg 0 \\ F^i = V_R, & i \ll 0 \end{cases}$

Morphism $f: (V, F) \longrightarrow (V', F')$ is a Λ-linear map

$f: V \longrightarrow V'$ s.t. $f(F^p V_R) \subset F'^p (V'_R)$

$$(V, F) \otimes (V', F') = (V \otimes V', F \otimes F')$$

$$(F \otimes F')^i (V \otimes V')_R = \sum_j F^j V_R \underset{R}{\otimes} F'^{i-j} V'_R$$

<u>Def</u> HS_Λ is the subcat of $\mathrm{Fil}_{\mathbb{C}/\Lambda}$ spanned by finite direct sums of pure Hodge str. of wt m.

<u>Fact.</u> HS_Λ is abelian, stable under tensor product.

<u>Rmk</u> $\mathrm{Fil}_{\mathbb{C}/\Lambda}$ is <u>NOT</u> abelian.

$\Lambda = \mathbb{R}$

<u>Lemma</u> Let $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$ $\quad (\mathbb{S}(\mathbb{R}) = \mathbb{C}^\times, \mathbb{S}(\mathbb{C}) = \mathbb{C}^\times \times \mathbb{C}^\times)$

$$\mathbb{S}(A) = (A \underset{\mathbb{R}}{\otimes} \mathbb{C})^\times$$

$HS_{\mathbb{R}} \simeq \mathrm{Rep}_{\mathbb{R}}(\mathbb{S})$ rep'ns of $\mathbb{S}$ on $\mathbb{R}$-v.s

$V = \oplus V^{p,q}_{\mathbb{C}} \rightsquigarrow [\, z$ acts on $V^{p,q}$ by $z^{-p} \bar{z}^{-q} \,]$

<u>Next time</u>. $HS_{\mathbb{Q}}$.

$$\mathbb{S} \curvearrowright V$$

$$\mathbb{S}_{\mathbb{C}} \curvearrowright V_{\mathbb{C}} = \oplus V^{p,q}_{\mathbb{C}}$$

Lecture 2. Λ Dedekind domain (eg. $\Lambda = \mathbb{Z}, \mathbb{Z}[\frac{1}{p}], \mathbb{Z}_{(p)}, \mathbb{Q}, \mathbb{R}$) $\subset \mathbb{R}$

$HS_\Lambda \subset \mathrm{Fil}_{\mathbb{C}/\Lambda}$ $\quad \{(V, F^\cdot) : F^i V_{\mathbb{C}} \supset F^{i+1} V_{\mathbb{C}} \supset \cdots + \cdots\}$

$\| $

$\bigoplus_m HS_{\Lambda, m} \leftarrow (V, F^\cdot)$ $\quad F^p \oplus \overline{F^{m-p+1}} = V_{\mathbb{C}}$ $\forall p$

$\uparrow$

Λ - Hodge str. pure of weight m $\Updownarrow$ $(V, F^\cdot) \leftrightarrow (V, V_{\mathbb{C}} = \bigoplus V_{\mathbb{C}}^{p,q}$, $V^{p,q} = \overline{V^{q,p}})$

Rmk $\mathrm{Fil}_{R/\Lambda}$ for general R, $\dfrac{F^i V_R}{F^{i+1} V_R}$ is finite proj - R-mod.

When $\Lambda = \mathbb{Q}, \mathbb{R}$, each $HS_{\Lambda, m}$ is abelian. $\overset{\Lambda = \mathbb{R}}{\overset{\otimes}{\longleftrightarrow}}$ Rep of $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$ $(V, h : \mathbb{S} \to GL(V_{\mathbb{R}}))$

For general Λ, HS_Λ is closed under $\otimes$.

Ex. X smooth proj. $/\mathbb{C}$

$$H^m_B(X^{an}, \Lambda)/\mathrm{tor}$$
$$\downarrow$$
$$(H^m_{dR}(X/\mathbb{C}), F^\cdot)$$

Subex. (1) X curve,

$$0 \to H^0(X, \Omega_X) \to H^1_{dR}(X/\mathbb{C}) \to H^1(X, \mathcal{O}_X) \to 0$$

(2) X abelian var. $/\mathbb{C}$, $e : \mathrm{Spec}\, \mathbb{C} \to X$

$$W = e^* \Omega^1_X = (\mathrm{Lie}\, X)^*, \quad \Omega^i_X = \mathcal{O}_X \otimes \Lambda^i W$$

$$0 \to (\mathrm{Lie}\, X)^* \to H^1_{dR}(X/\mathbb{C}) \to \mathrm{Lie}\, X^\vee \to 0$$

$$\begin{tikzcd} & & H_1(X) \arrow[d] & & \\ 0 \arrow[r] & (\mathrm{Lie}\, X^\vee)^* \arrow[r] & H_1^{dR}(X/\mathbb{C}) \arrow[r] & \mathrm{Lie}\, X \arrow[r] & 0 \end{tikzcd}$$

$$\mathbb{G}_m \xrightarrow{w} \mathbb{S} \curvearrowright V$$

$$\mathbb{R}^\times \longrightarrow \mathbb{C}^\times$$

$$r \longmapsto r^{-1}$$

(V, F) is of weight m iff $w(\mathbb{G}_m)$ acts as $r \longmapsto r^m$.

Ex $\begin{pmatrix} \mathbb{S} \longrightarrow \mathbb{G}_m \\ z \longmapsto |z|^{2m} \end{pmatrix} \longleftrightarrow$ real Hodge str. $\mathbb{R}(m)$

$\left((2\pi i)^m \mathbb{R} = V, \quad V_{\mathbb{C}} = V_{\mathbb{C}}^{-m,-m} \right)$

$\dim_{\mathbb{C}} X = d$,

$$\begin{tikzcd} (2\pi i)^d \Lambda \arrow[r, phantom, "\simeq"] \arrow[d, hook] & H_B^{2d}(X^{an}, \Lambda) \arrow[d, hook] \\ \mathbb{C} & H_{dR}^{2d}(X/\mathbb{C}) \arrow[l, "\simeq"'] \end{tikzcd}$$

$$\text{is}$$

$$H_{dR}^d(X, \Omega^d)$$

choose $i \rightsquigarrow$ orientation of X^{an}

$\rightsquigarrow [X^{an}]$

Polarization

Def. A polarization on a pure Hodge str. (V, h) of wt m is a Λ-bilinear map

$Q : V \otimes V \longrightarrow \Lambda(-m)$ s.t.

$B_Q : \quad V_{\mathbb{R}} \times V_{\mathbb{R}} \longrightarrow \mathbb{R}$

$$B_Q(v,w) = (2\pi i)^m Q_{\mathbb{R}}(v, h(i)w) \quad \text{is symmetric and pos. def.}$$

<u>Lemma</u> Q is non-degenerate (Λ field) and $(-1)^m$-sym.

<u>Pf.</u> Non-degeneracy is clear.

On $\Lambda(-m)$, $h(z) = |z|^{-2m}$

$$\begin{aligned}
Q_{\mathbb{R}}(v,w) &= h(i) \cdot Q_{\mathbb{R}}(v,w) \\
&= Q_{\mathbb{R}}(h(i)v, h(i)w) \\
&= (2\pi i)^{-m} B_Q(h(i)v, w) \\
&= (2\pi i)^{-m} B_Q(w, h(i)v) \\
&= Q_{\mathbb{R}}(w, h(-1)v) \qquad \mathbb{G}_m \xrightarrow{w} \mathbb{S} \xrightarrow{h} GL(V_{\mathbb{R}}) \\
&= (-1)^m Q_{\mathbb{R}}(w,v)
\end{aligned}$$

<u>Rmk</u> We can extend B_Q to a Hermitian form

$$H_Q : V_{\mathbb{C}} \otimes \overline{V_{\mathbb{C}}} \longrightarrow \mathbb{C}$$

$$v \otimes w \mapsto (2\pi i)^n Q_{\mathbb{C}}(v, h(i)\bar{w})$$

Then Q is a polarization $\Leftrightarrow$
(1) $H_Q(V_{\mathbb{C}}^{p,q}, \overline{V_{\mathbb{C}}^{p',q'}}) = 0$ unless $p = q'$, $q = p'$
(2) $H_Q|_{V^{p,q}}$ is positive def.

Hodge-Riemann bilinear relations

We say $(V, h) = \bigoplus_m (V_m, h)$ is polarizable if each (V_m, h) admits a polarization.

$PHS_{\Lambda, m} \subset HS_{\Lambda, m}$ polarizable Hodge str. (of wt m)

$\underbrace{PHS_\Lambda}_{\text{closed under } \otimes} \subset HS_\Lambda$

Ex X smooth proj. $/\mathbb{C}$, $\omega \in H^2(X^{an}, \mathbb{Z})(1)$

$\omega = c_1(\mathcal{L})$, $\mathcal{L}$ ample

$$1 \to (2\pi i)\mathbb{Z} \to \mathcal{O} \xrightarrow{\exp} \mathcal{O}^\times \to 1$$

$$L^j : H^i(X^{an}, \mathbb{Q}) \xrightarrow{\cup \omega^j} H^{i+2j}(X^{an}, \mathbb{Q}(j))$$

is a map of pure Hodge str. of wt i.

Let $P^i = \ker(L^{n-i+1} : H^i \to H^{2n-i+2})$ ← primitive

$$H^i \xrightarrow{\omega^{n-i}} H^{2n-i}(\cdot\cdot) \to H^{2n-i+2}$$

$$\cup$$

$$P^i \xrightarrow{L^{n-i}} L^{n-i}(P^i)(\cdot\cdot)$$

$$H^i = \bigoplus_{j \geq 0} L^j P^{i-2j}(-j)$$

$$\begin{array}{ccccc} P^i & \otimes & P^i & & \mathbb{Q}(-n) \\ \downarrow & & \downarrow L^{n-i} & & \| \\ H^i & \otimes & H^{2n-i}(n-i) & \longrightarrow & H^{2n}(n-i) \simeq \mathbb{Q}(-i) \end{array}$$

Thm $(\cdot,\cdot) : P^i \otimes P^i \to \mathbb{Q}(-i)$ is a polarization of P^i.

Cor X smooth proj, $H^m(X, \mathbb{Q}) \in PHS_{\mathbb{Q}, m}$

Rmk. Hodge decomp. exists for coh. of cpt Kähler mfds. X

But $H^m(X, \mathbb{Q})$ is in general not $\mathbb{Q}$-polarizable.

Thm There is an equiv. of cats

$$\left\{ \text{abelian var.}/\mathbb{C} \right\} \simeq \left\{ \begin{array}{l} \mathbb{Z}\text{-polarizable Hodge str. } (V, h) \text{ of} \\ \text{wt } -1 \text{ and } V_{\mathbb{C}} = \bigoplus V_{\mathbb{C}}^{p,q} \\ V^{p,q} = 0 \text{ unless } (p,q) = (-1,0), (0,-1) \end{array} \right\}$$

$$A \longmapsto (H_1(A;\mathbb{Z}), F)$$

$$A = V_{\mathbb{C}}^{0,-1}/V \longleftarrow\!\shortmid (V, h)$$

$$\begin{tikzcd} V \arrow[r] \arrow[dr] & V_{\mathbb{C}} = V_{\mathbb{C}}^{-1,0} \oplus V_{\mathbb{C}}^{0,-1} \arrow[d] \\ & V_{\mathbb{C}}^{0,-1} \end{tikzcd}$$

$[E : \mathbb{Q}] = 2g$

Ex Let E be a CM field w $*: E \to E$ complex conj. Let $V = E$, regarded as a $\mathbb{Q}$-v.s. We choose a CM type $\Phi \subset \mathrm{Hom}(E, \mathbb{C})$

$$\Phi \sqcup *(\Phi) = \mathrm{Hom}(E, \mathbb{C})$$

$$\iota: V_{\mathbb{R}} \xrightarrow{\sim} \mathbb{C}^{\Phi}$$

$$(v \otimes \iota) \mapsto \prod_{\sigma \in \Phi} \iota\, \sigma(v)$$

$$h: \mathbb{S} \longrightarrow GL(V_{\mathbb{R}})$$

$z \curvearrowright \mathbb{C}^{\Phi}$ by scalar

$(V = E, h)$ is a pure $\mathbb{Q}$-Hodge str. of weight -1.

Choose $f \in E$ s.t. $f^* = -f$. Fix i.

and $i\sigma(f) > 0$, $\forall \sigma \in \Phi$

Then $Q_f : V \otimes V \to \mathbb{Q}(1)$

$$v, w \mapsto (2\pi i)\, \mathrm{Tr}_{E/\mathbb{Q}}(f v w^*)$$

is a polarization.

$(V = E, \Phi, f)$ polarized $\mathbb{Q}$-HS

$\mathfrak{a} \subset \mathcal{O}_E$ order, $(\mathfrak{a}, \Phi) \longleftrightarrow A_{\mathfrak{a}, \Phi}$ $\quad A = \mathbb{C}^{\Phi} / \iota(\mathfrak{a})$

CM abelian var.

when $\mathfrak{a} = \mathcal{O}_E$, $A_\Phi = A_{\Phi, \mathcal{O}_E}$.

<u>Lecture 3</u> $(V,h) \in$ $h : \mathbb{S} \to GL(V_{\mathbb{R}})$

$HS_\Lambda \supset PHS_\Lambda$

$\Lambda = \mathbb{R}$, $HS_{\mathbb{R}} \simeq \mathrm{Rep}_{\mathbb{R}}(\mathbb{S})$

$HS_{\mathbb{Q}} \ni (V,h)$

<u>Def.</u> The Mumford–Tate group of (V,h) is the smallest $\mathbb{Q}$-closed subgroup of $GL(V)$ containing $h(\mathbb{S}) \subset GL(V_{\mathbb{R}})$, denoted by $G_{MT}(V,h)$ or $G_{MT}(V)$ or G_{MT}.

<u>Prop</u> Write $G = G_{MT}(V,h)$

(1) G is conn'd (in Zar. top.)

(2) $G_{MT}(V_1 \oplus V_2) \subset GL(V_1) \times GL(V_2) \subset GL(V_1 \oplus V_2)$

& the projection $G_{MT}(V_1 \oplus V_2) \subset GL(V_1) \times GL(V_2)$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ G_{MT}(V_i) & \subset & GL(V_i) \end{array}$$

(3) $\exists$ a tautological $h: \mathbb{S} \longrightarrow G_{\mathbb{R}}$ s.t. $w_h := h \circ w : \mathbb{G}_m \longrightarrow G_{\mathbb{R}}$ is defined $/\mathbb{Q}$

In addition, $w_h(\mathbb{G}_m) \subset Z_G$ (center of G)

[Recall $w: \mathbb{G}_m \longrightarrow \mathbb{S}$, $z \longmapsto z^{-1}$]

(4) $\exists$ a nat'l fully faithful $\mathbb{Q}$-linear exact $\otimes$-functor

$$\mathrm{Rep}_{\mathbb{Q}}(G) \longrightarrow HS_{\mathbb{Q}}$$
$$(V, \rho) \longmapsto (V, \rho_{\mathbb{R}} \circ h)$$

w/ essential image closed under taking $\otimes$, $\oplus$, dual, subobj., qt obj.

(5) If V is polarizable, ~~then~~. Let $G_h = \{ g \in G(\mathbb{C}) : h(i) \bar{g} h(i)^{-1} = g \}$

Then $G_h / w(\mathbb{G}_{m\,\mathbb{R}})$ is compact.

(i.e. $\mathrm{Ad}\, h(i) : \left(G/w(\mathbb{G}_m)\right)_{\mathbb{R}} \longrightarrow \left(G/w(\mathbb{G}_m)\right)_{\mathbb{R}}$ is a Cartan involution of $\left(G/w(\mathbb{G}_m)\right)_{\mathbb{R}}$)

<u>Sketch of proof</u> (1), (2) ✓

(3): $V = \bigoplus V_i$, V_i wt m_i, $w_h(z)$ acts as z^{m_i} on V_i, so

w_h is defined $/\mathbb{Q}$

$$w_h(\mathbb{G}_m) \subset \prod Z_{GL(V_i)} \Rightarrow w_h(\mathbb{G}_m) \subset Z_G$$

(4) Need to show $\mathrm{Hom}_G(W_1, W_2) \xrightarrow{\sim} \mathrm{Hom}_{HS_{\mathbb{Q}}}(W_1, W_2)$

By taking dual, reduce to show $\mathrm{Hom}_G(\mathbb{Q}, W) \xrightarrow{\sim} \mathrm{Hom}_{HS_{\mathbb{Q}}}(\mathbb{Q}, W)$

$$\begin{array}{ccc} \| & & \| \\ W^G & & W^{\rho_{\mathbb{R}} \circ h(\mathbb{S})} \\ & & \| \\ & & W \cap W_{\mathbb{R}}^{\rho_{\mathbb{R}} \circ h(\mathbb{S})} \end{array}$$

Let $H = \{ g \in GL(V) : g.v = v, \quad \forall v \in W \cap W_{\mathbb{R}}^{\rho_{\mathbb{R}} \circ h(\mathbb{S})} \}$

$H_{\mathbb{R}} \supset \rho_{\mathbb{R}} \circ h(\mathbb{S}) \quad \Rightarrow \quad H \supset G$

$\Rightarrow \quad W^G \supset W^H \supset W \cap W_{\mathbb{R}}^{\rho_{\mathbb{R}} \circ h(\mathbb{S})}$

(5) Can assume V is pure of wt m.

$Q : V \otimes V \longrightarrow \mathbb{Q}(-m)$
$\neq$
0

$\mathbb{Q}(-m)$ is in the essential image of $\mathrm{Rep}_{\mathbb{Q}}(G) \to HS_{\mathbb{Q}}$.

(4.5) Let $\langle V \rangle^{\otimes}$ be the subcat of $HS_{\mathbb{Q}}$ containing (V, h)

& is closed under taking $\otimes$, dual, $\oplus$, subobj, quotient obj.

Then $\mathrm{Rep}(G) \xrightarrow{\sim} \langle V \rangle^{\otimes} \subset HS_{\mathbb{Q}}$.

Lemma Let V be a f.d. vec. sp. / field k.

Let $H \subset GL(V)$ be a closed subgp. Then every rep'n of H appears as a subquotient of finite direct sums of $\{V^{\otimes r} \otimes (V^\vee)^{\otimes s}\}_{r,s}$

$\overset{(4)}{\Longrightarrow}$ (4.5)

Recall (4) $\mathrm{Rep}(G) \longrightarrow HS_{\mathbb{Q}}$

$(W, \rho) \longmapsto (W, \rho_{\mathbb{R}} \circ h)$

exact, $\otimes$, $\oplus$, fully faithful essential image stable under taking $\otimes$, $\oplus$, sub. qt.

$W \in \mathrm{Rep}(G) \rightsquigarrow (W, \rho_{\mathbb{R}} \circ h)$

Suppose $W' \subset W$ sub Hodge str.

$$\begin{tikzcd}
 & (\rho_{W' \subset W})_{\mathbb{R}} \arrow[dl] \arrow[dr] & \\
GL(W') & & GL(W_{\mathbb{R}}) \\
 & G_{\mathbb{R}} \arrow[ul, dashed] \arrow[ur] \arrow[uu, leftarrow] & \\
 & \mathbb{S} \arrow[u] \arrow[ul, leftarrow] \arrow[uu, bend right, leftarrow] &
\end{tikzcd}$$

Pf (5) Suppose V pure of wt m

$Q : V \otimes V \longrightarrow \mathbb{Q}(-m)$, $Q \neq 0$

$\Rightarrow \mathbb{Q}(-m) \longleftrightarrow \chi : G \to \mathbb{G}_m$

$Q(gv, gw) = \chi(g)\, Q(v, w)$

Let $H_Q : V_{\mathbb{C}} \times \overline{V_{\mathbb{C}}} \longrightarrow \mathbb{C}$ be the Hermitian form assoc. to Q.

$H_Q(v, w) = (2\pi i)^m Q_{\mathbb{C}}(v, h(i)\overline{w})$

Let $k \in G_h$, $H_{\mathbb{Q}}(kv, kw) := (2\pi i)^m Q_{\mathbb{C}}(kv, h(i)\bar{k}\bar{w})$

$$= (2\pi i)^m Q_{\mathbb{C}}(kv, k\, h(i)\bar{w})$$

$$= \chi(k) H_{\mathbb{Q}}(v, w)$$

$\Rightarrow G_h \subset GU(V, H_{\mathbb{Q}})$

Cor. $\chi : G \to G_m \longleftrightarrow \mathbb{Q}(m)$ for some m.

χ is trivial $\Leftrightarrow m = 0$.

<u>Notation</u> Let V be a finite free (projective) Λ-module.

$$T^{\bullet}(V) = \bigoplus_{r \geq 0} V^{\otimes r}, \quad T^{\bullet,\bullet}(V) = \bigoplus_{r,s \geq 0} V^{\otimes r} \otimes (V^{\vee})^{\otimes s}$$

algebras

$$= T^{\bullet}(V) \otimes T^{\bullet}(V^{\vee}) \supset \bigoplus_{r \geq 0} (V \otimes V^{\vee})^{\otimes r}$$

<u>Def</u> A <u>Hodge tensor</u> of type (p,p) is an elt in $T^{\bullet,\bullet}(V) \cap T^{\bullet,\bullet}(V)^{p,p}_{\mathbb{C}}$

A <u>Hodge cycle</u> is a Hodge tensor of type $(0,0)$

<u>Rmk</u> If V is pure of wt m, $b \in T^{r,s}(V) = V^{\otimes r} \otimes (V^{\vee})^{\otimes s}$ is a Hodge tensor of type (p,p). Then $2p = (r-s)m$.

If $m \neq 0$ $\Longrightarrow$ Hodge cycles $\subset \bigoplus_{r \geq 0} T^{r,r}$

Let $C^{\cdot,\cdot}_{Hdg}(V) \subset T^{\cdot,\cdot} V$ be the subalg of Hodge cycles.

If V is pure of wt $m \neq 0$, we write it as $C^{\cdot}_{Hdg}(V) \subset \bigoplus_{r \geq 0} (V \otimes V^\vee)^{\otimes r}$

Ex. $\mathrm{End}\, V \simeq V \otimes V^\vee$

$$\begin{array}{c} \cup \\ B := \mathrm{End}_{HS_\Lambda} V \simeq C^{1,1}_{Hdg}(V) \end{array}$$

$$B_r := \mathrm{End}_{HS_\Lambda}(V^{\otimes r}) \simeq C^{r,r}_{Hdg}(V).$$

Let $C^{\cdot}_{Hdg}(V)^1 \subset C^{\cdot}_{Hdg}(V)$ be the subalg. gen. by $C^{1,1}_{Hdg}(V)$

$$B_1 \twoheadleftarrow C^{\cdot}_{Hdg}(V)^1 \subset C^{\cdot}_{Hdg}(V)$$
$$\cap$$
$$C^{\cdot,\cdot}_{Hdg}(V)$$

Thm Let $(V, h) \in HS_{\mathbb{Q}}$. The following subgp of $GL(V)$ are the same

(1) $G_{MT}(V, h)$

(2) $\{ g \in GL(V) : gb = b, \ \forall b \in C^{\cdot,\cdot}_{Hdg}(V) \}$

(3) The tannakian group of $\langle (V,h) \rangle^{\otimes} \subset HS_{\mathbb{Q}}$.

Rmk If $C^{\cdot}_{Hdg}(V)^1 = C^{\cdot,\cdot}_{Hdg}(V) \Rightarrow G = GL_{B_1}(V)$.

$\mathrm{Rep}_k(H)$ abelian, $\otimes$, $(-)^\vee \xrightarrow[\text{functor}]{\text{faithful } \otimes\text{-exact}} \mathrm{Vect}_k$

Lecture 4 **Thm** Let $(V,h) \in HS_{\mathbb{Q}}$. Then the following closed $\mathbb{Q}$-subgp of $GL(V)$ are the same

(1) $G_{MT}(V,h) =: G$

(2) $\{ g \in GL(V) : gb = b, \ b \in \underset{\substack{\| \\ (T^\bullet V \otimes T^\bullet V^\vee)^{h(\mathbb{S})}}}{C^{\bullet,\bullet}_{Hdg}(V)} \} =: G'$

(3) Tannakian group of $\langle V,h \rangle^{\otimes} \subset HS_{\mathbb{Q}}$

full subcat containing (V,h) closed under $\otimes, \oplus, (-)^\vee$, sub, qt.

$$\left(\Leftrightarrow \begin{tikzcd} \mathrm{Rep}_{\mathbb{Q}}(G) \arrow[r] \arrow[dr, "\sim"'] & HS_{\mathbb{Q}} \\ & \langle V,h \rangle^{\otimes} \arrow[u] \end{tikzcd} \right)$$

(1) = (3). follows from

Lemma Let V be a f.d. v.s. / k field, $H \subset GL(V)$ closed subgp. then every rep of H appears as a subquotient of a finite direct sum of $\{ V^{\otimes r} \otimes (V^\vee)^{\otimes s} \}_{r,s \in \mathbb{Z}_{\geq 0}}$

Pf. $H \times W \longrightarrow W \iff W \longrightarrow k[H] \otimes \underline{W}$ H-equiv. map

injective

$\| \qquad$ H acts on $k[H]$ via left translation

$k[H]^{\oplus \dim \underline{W}}$ on $\underline{W}$ by trivial action.

Enough to show that every finite dim'l subrep of $k[H]$ appears as a subqt of a finite direct sum of $\{ V^{\otimes r} \otimes (V^\vee)^{\otimes s} \}_{r,s}$

$$\mathrm{Sym}(\underset{\substack{\| \\ V \otimes \underline{V^\vee}}}{\mathrm{End}(V)})[\det^{-1}] \longrightarrow k[H] \qquad (H \subset GL(V) \text{ closed subgp})$$

$T^*V \otimes T^*V^\vee \longrightarrow \mathrm{Sym}(V \otimes \underline{V^\vee})$

$\det V \hookrightarrow V^{\otimes n}$, $n = \dim V$

$e_1 \wedge \cdots \wedge e_n \mapsto \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sgn}(\sigma)\, e_{\sigma(1)} \otimes \cdots \otimes e_{\sigma(n)}$

$(V^\vee)^{\otimes n} \longrightarrow (\det V)^\vee$

(1) = (2) : Clearly $G \subset G'$ ✓

<u>Lemma</u> Let $H_1 \subset H_2 \subset GL(V)$ closed subgps $/k$ field (there's a more general statement for k = Dedekind domain)

Then (1) $\exists$ f.d. rep W_2 of H_2, contained in

finite direct sum $\{V^{\otimes r} \otimes (V^\vee)^{\otimes s}\}_{r,s}$ & $W_1 \subset W_2$ 1-dim'l subspace

s.t. $H_1 = \{h \in H_2 : h(W_1) = W_1\}$

(2) Let $H' = \{h \in H_2 : ht = t, \ \forall t \in T^{\cdot\cdot}(V)^{H_1}\}$.

If for every 1-dim. rep L of H_1, $\exists\, a \in \mathbb{Z}_{>0}$ s.t. L^a can be H_1-equiv. embedded into $T^{\cdot\cdot}(V)$, then $H_1 = H'$.

$\ulcorner$ Applying Lemma (2): $H_1 = G$, $V = \bigoplus_i V_i$, $H_1 \subset H_2 = \prod_i GL(V_i) \subset GL(V)$, $H' = G'$

1-dim'l rep of $G \longleftrightarrow \mathbb{G}(m)$

$\boxed{\text{Case 1}}$

If $V = V_1$ pure of wt 0, every obj. in $\langle V^{\otimes} \rangle$ is of wt 0

$\Rightarrow m = 0$, i.e. Every 1-dim'l rep of G is trivial.

Rogers

$\therefore$ can be G-equiv. embedded into $T^{\cdot,\cdot}(V)$.

$\boxed{\text{Case 2}}$ $\quad V = \bigoplus V_i, \quad wt(V_1) = m \neq 0$.

$$\det V_1 = \mathbb{C}\Big(\underbrace{-\frac{m_1 \dim V_1}{2}}_{\neq 0}\Big)$$

$$\mathbb{C}(m)^{\otimes a} = (\det V_1)^b \quad \text{for some } a, b$$

$\Rightarrow \mathbb{C}(m)^{\otimes a}$ embedded into a finite direct sum of $V^{\otimes r} \otimes (V^\vee)^{\otimes s}$.

<u>Pf of Lemma</u>

(1) Can assume $H_2 = GL(V)$

$$\begin{tikzcd} H_1 \arrow[r, hook] \arrow[d] & GL(V) \arrow[d, phantom, "\cap"] \\ \bar{H}_1 \arrow[r, "\text{closed}"] & \mathrm{End}(V) \end{tikzcd} \qquad \bar{H}_1 \longleftrightarrow \begin{tikzcd} I \arrow[r, phantom, "\subset"] & \mathrm{Sym}(V \otimes V^\vee) \\ \tilde{I} \arrow[r, phantom, "\subset"] \arrow[u, two heads] & T^{\cdot,\cdot}(V) \arrow[u, two heads] \end{tikzcd}$$

$\tilde{I}$ is a two-sided ideal in $T^{\cdot,\cdot}(V)$

$$\bar{H} = \{ g \in \mathrm{End}(V) : \bar{H} g \subset \bar{H} \} \qquad H = \bar{H} \cap GL(V)$$

$$\Leftrightarrow H = \{ g \in GL(V) : a_r(g) I \subset I \},$$

a_r action of $GL(V)$ on $k[\mathrm{End}\, V]$ by right translation

$$= \{ g \in GL(V) : g\tilde{I} \subset \tilde{I} \}$$

Pick set of generators of $\tilde{I}$, choose $U_2 \subset T^{\cdot,\cdot}(V)$ finite dim'l rep of $GL(V)$ containing the generators. $U_1 = U_2 \cap \tilde{I}$. $\Rightarrow$ set of generators of $\tilde{I}$

$\tilde{I}$

$\cup$

$U_1 \subset U_2$

$\{e_1, \dots, e_r\} \subset \{e_1, \dots, e_n\}$

$\det U_1 \subset U_1^{\otimes r} \subset U_2^{\otimes r}$

$\| $

W_1

$\sum \mathrm{sgn}(\sigma) e_{\sigma(i)} \qquad e_{i_1} \otimes \dots \otimes e_{i_r}$

(2) $H_1 \subset H' \subset H_2$

Take $W_1 \subset W_2$ as in (1).

Let $L = W_1^\vee$, $L^{\otimes a} \subset N \subset T^{\cdot,\cdot}(V)$, N finite dim'l rep of H_2

$H_1 = \{ h \in H_2 : hv = v, \ v \in W_1^a \otimes L^a \subset W_2^{\otimes a} \otimes N \subset$ finite sum of $V^{\otimes r} \otimes (V^\vee)^{\otimes s} \}$

Note H' fixes $v \in W_1^a \otimes L^a$, $H' \subset H_1$. $\square$

<u>Rmk</u> (1) $B_1 := \mathrm{End}_{HS_{\mathbb{Q}}}(V, h) \twoheadleftarrow C_{Hdg}(V)^1 \subset C_{Hdg}^{\cdot}(V) \subset C_{Hdg}^{\cdot,\cdot}(V)$

$\uparrow$ generated by $C_{Hdg}^{1,1}(V)$

$G_{MT}(V, h) \subset GL_{B_1}(V)$

equality holds if V pure of wt $m \neq 0$, and $C_{Hdg}^{\cdot,\cdot}(V)$ is gen. by $C_{Hdg}^1(V)$.

(2) If $(V, h) \in PHS_{\mathbb{Q}}$, $G_{MT}(V, h)$ is reductive b/c $(G/W_h(\mathbb{G}_m))_{\mathbb{R}}$ has a Cartan involution

$\Rightarrow PHS_{\mathbb{Q}} = \bigcup \langle V, h \rangle^{\otimes}$ is semisimple abelian tensor cat.

(3) If (V, h) is polarizable, irreducible,

$D := \mathrm{End}_{HS_{\mathbb{Q}}}(V, h)$ f.d. division algebra $/\mathbb{Q}$.

In addition, let $Q : V \otimes V \longrightarrow \mathbb{Q}(-m)$ be a polarization.

$* : D \longrightarrow D$, $\phi \longmapsto \phi^*$ s.t. $Q(\phi v, w) = Q(v, \phi^* w)$, $\forall\, v, w \in V$

$\phi^* = Q^{-1} \circ \phi^\vee \circ Q$, Q regarded as $V \longrightarrow V^\vee(-m)$.

<u>Lemma</u> $*$ is positive. I.e. $\mathrm{Tr}(\phi^* \phi|_V) > 0$ for $\phi \neq e$

<u>Pf</u> Need to show $\mathrm{Tr}(\phi^* \phi|_{V_{\mathbb{R}}}) > 0$. $B_Q : V_{\mathbb{R}} \otimes V_{\mathbb{R}} \longrightarrow \mathbb{R}$ positive definite.

$$B_Q(v, \phi^* \phi v) \overset{h \circ \phi = \phi \circ h}{=} B_Q(\phi v, \phi v) > 0, \quad v \neq 0$$

$$\Rightarrow \mathrm{Tr}(\phi^* \phi|_{V_{\mathbb{R}}}) > 0.$$

$\exists$ a complete classification of isom. classes of $(D, *)$ positive def. inv. by Albert ?

<u>Ex</u> If $E = D$ is commutative so it's a number field.

Then E is either a totally real field ($* = \mathrm{id}$) or a CM field ($*$ = cplx conj.) $\leftarrow$ later, we can both totally real field & CM field as CM field.

<u>Ex</u> (1) $G_{MT}(\mathbb{Q}(m)) = \begin{cases} * & , m = 0 \\ \mathbb{G}_m & , m \neq 0 \end{cases}$

(2) E quadratic imaginary, $E \hookrightarrow \mathbb{C}$

$(E, h) \rightsquigarrow G_{MT} = \mathrm{Res}_{E/\mathbb{Q}} \mathbb{G}_m$, $E^\times \curvearrowright E$

(3) In general, $\dim V = 2$, $V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1}$

$$G_{MT}(V) = \begin{cases} \mathrm{Res}_{E|\mathbb{Q}} \mathbb{G}_m \\ GL(V) \end{cases}$$

(V, h) polarizable, $\dim V = 2g$, $V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1}$

$$G_{MT} \subset GSp_{2g}(V)$$

<u>Lecture 5</u> <u>Def</u>. A CM $\mathbb{Q}$-Hodge structure is a polarizable pure $\mathbb{Q}$-Hodge str. (V, h)

s.t. $G_{MT}(V, h)$ is a torus.

$CMHS_{\mathbb{Q}} \subset PHS_{\mathbb{Q}} \subset HS_{\mathbb{Q}}$.

$\otimes$-closed

<u>Thm</u>. $CMHS_{\mathbb{Q}} \simeq \mathrm{Rep}_{\mathbb{Q}}(\mathfrak{S})$ ← protorus, called the Serre group

<u>Ex</u>. (1) Let $E|F$ be a CM field, $F = E^{*=1}$. $\Phi \subset \mathrm{Hom}(E, \mathbb{C})$ CM type

$\rightsquigarrow (V = E, h)$ wt -1, $G_{MT}(V, h) \subset \mathrm{Res}_{E|\mathbb{Q}}(\mathbb{G}_m)$

$V \otimes \mathbb{R} \simeq \mathbb{C}^{\Phi}$

torus $\cap$ $\subset$

$\mathbb{G}\mathbb{U}_{E|F}(1) = \{ t \in E^{\times} : \mathrm{Nm}_{E|F}(t) \in \mathbb{Q}^{\times} \}$

(0) $\mathbb{Q}(m) \in CMHS_{\mathbb{Q}}$, $G_{MT} = \begin{cases} \mathbb{G}_m, & m \neq 0 \\ *, & m = 0 \end{cases}$

(2) Let (E, Φ_i) be a finite collection of CM types of E (A CM type could appear w/ multiplicity in this collection)

$$\bigotimes_i (V_i, h_i) \circlearrowleft \bigotimes_i E =: A$$

$$\bigotimes_i (V_i, h_i)^{\varepsilon_i} \underset{\bigotimes_i E}{\otimes} E =: (V, h) \in CMHS_{\mathbb{Q}}$$

$\varepsilon_i = \{\pm 1\}$

wt $-\left(\sum \varepsilon_i\right)$

$\xrightarrow{\sim}$

Some remarks on alg. tori

Let k be a field, T/k a torus

$X_*(T) = \underline{\mathrm{Hom}}(\mathbb{G}_m, T)$

$X^*(T) = \underline{\mathrm{Hom}}(T, \mathbb{G}_m)$

$\left\{ \begin{array}{c} \Omega \text{ alg closed field} \\ \text{ext'ns of } k \end{array} \right\} \longrightarrow$ finite free abelian gps

$$\Omega \longmapsto \mathrm{Hom}(\mathbb{G}_{m,\Omega}, T_\Omega) \circlearrowleft \mathrm{Aut}(\Omega/k)$$

$$\mathrm{Hom}(T_\Omega, \mathbb{G}_{m,\Omega}) \circlearrowleft \mathrm{Aut}(\Omega/k)$$

E/k finite sep. ext'n

$R_{E/k} := \mathrm{Res}_{E/k}\, \mathbb{G}_m$, $k \hookrightarrow \Omega$, $(R_{E/k})(\Omega) = (E \otimes_k \Omega)^\times$

$\mathrm{Aut}(E/k) \overset{\text{free}}{\circlearrowright} \mathrm{Hom}_k(E, \Omega) \overset{\text{transitive}}{\circlearrowleft} \mathrm{Aut}(\Omega/k)$

$$X^*(R_{E/k}) = \mathbb{Z}^{\mathrm{Hom}_k(E,\Omega)}$$

$$= \left\{ \sum n_\sigma [\sigma] : \sigma \in \mathrm{Hom}_k(E, \Omega) \right\}$$

$\{\mu_\sigma\}$ dual basis

<u>Notation</u>: $f : \mathrm{Hom}(E, \Omega) \to \mathbb{Z}$, $\mu_f = \sum f(\sigma) \mu_\sigma$

<u>Rmk</u>: E/k Galois, $R_{E/k} \circlearrowleft \mathrm{Gal}(E|k)$ as k-torus

$R_{E/k}$ is equipped w/ $\mu^{E/k} : (\mathbb{G}_m)_E \longrightarrow (R_{E/k})_E$

$$R^\times \longmapsto (R \otimes_k E)^\times$$

Given $\sigma : E \to \Omega$, $(\mu^{E/k})_\Omega = \mu_\sigma$.

$\mathbb{G}_m \xrightarrow{\omega} R_{E/k}$ defined $/k$

$R^\times \longrightarrow (R \otimes_k E)^\times$

$r \longmapsto r^{-1} \otimes 1$

$E'|E|k$, $N_{E'|E} : R_{E'/k} \longrightarrow R_{E/k}$

$$N_{E'/E} \cdot \mu^{E'/k} = (\mu^{E/k})_{E'}$$

$\bar{k}$ sep. closure of k, $\left(R_{\bar{k}/k} := \varprojlim_{E \subset \bar{k}} R_{E/k}, \ \mu^{\bar{k}/k} \right)$

<u>Lemma</u> In the cat. of k-tori, the functor $T \longmapsto \mathrm{Hom}_E(\mathbb{G}_{m,E}, T_E)$ is corep. by $(R_{E/k}, \mu^{E/k})$

$$\mathrm{Hom}_{k\text{-tori}}(R_{E/k}, T) = \mathrm{Hom}_E(\mathbb{G}_{m,E}, T_E)$$

$$\varphi \longmapsto \varphi_E \cdot \mu^{E/k}$$

In particular, $T \mapsto \mathrm{Hom}_{\bar{k}}(\mathbb{G}_{m,\bar{k}}, T_{\bar{k}})$ is pro-corepresented by

$(R_{\bar{k}/k}, \mu^{\bar{k}/k})$.

<u>Ex</u> $k = \mathbb{R}$, $\bar{k} = \mathbb{C}$

$$\mathrm{Hom}(\mathbb{S}, T) = \mathrm{Hom}(\mathbb{G}_{m\mathbb{C}}, T_{\mathbb{C}})$$

$$\mathbb{G}_{m\mathbb{C}} \xrightarrow{\mu^{\mathbb{C}/\mathbb{R}}} \mathbb{S}_{\mathbb{C}} \simeq \mathbb{G}_{m\mathbb{C}} \times \mathbb{G}_{m\mathbb{C}}$$

$$(R \otimes \mathbb{C})^\times \to R^\times \times R^\times$$

$$(r \otimes z) \mapsto (rz, r\bar{z})$$

$\mu^{\mathbb{C}/\mathbb{R}}$ is the inclusion of the first factor, usually called the Hodge cocharacter

$w: \mathbb{G}_m \to \mathbb{S}$ is the weight cocharacter we introduced before.

$$\{h: \mathbb{S} \to T\} \leftrightarrow \{\mu_h : \mathbb{G}_{m\mathbb{C}} \to T_{\mathbb{C}}\}$$

$$\mu_h = h_{\mathbb{C}} \circ \mu^{\mathbb{C}/\mathbb{R}}$$

<u>Ex.</u> Let E be a # field, fix $\sigma_0 : E \to \mathbb{C}$.

$$\mu_{\sigma_0} : \mathbb{G}_{m\mathbb{C}} \to (R_{E/\mathbb{Q}})_{\mathbb{C}} \leftrightarrow h_{\sigma_0} : \mathbb{S} \to (R_{E/\mathbb{Q}})_{\mathbb{R}}$$

$$\mathbb{C}^\times \to (E \otimes \mathbb{R})^\times \simeq \prod \mathbb{R}^\times \times \prod \mathbb{C}^\times$$

$$h_{\sigma_0}(z) = \begin{cases} (z\bar{z}, 1, \cdots 1) & \text{if } \sigma_0 \text{ is real} \\ (z, 1, \cdots 1), & \text{if } \sigma_0 \text{ is complex} \end{cases}$$

The first component is induced by σ_0

$k = \mathbb{Q}$, T $\mathbb{Q}$-torus, $\Omega = \mathbb{C}$

$$\mathrm{Hom}(\mathbb{G}_{m\mathbb{C}}, T_{\mathbb{C}}) = \mathrm{Hom}(\mathbb{S}, T_{\mathbb{R}})$$

Consider the functor $\mathbb{Q}$-tori to (finite free) abelian groups

$$T \mapsto \{ h : \mathbb{S} \to T_{\mathbb{R}} : \mathbb{G}_m \xrightarrow{w} \mathbb{S} \xrightarrow{h} T_{\mathbb{R}} \text{ is defined over } \mathbb{Q} \}$$

$$\overset{\overset{w_h}{\frown}}{}$$

$$\begin{array}{c} \| \\ \mathrm{Hom}^0(\mathbb{S}, T_{\mathbb{R}}) \subset \mathrm{Hom}(\mathbb{S}, T_{\mathbb{R}}) \end{array}$$

Note that under $\mathrm{Hom}(\mathbb{S}, T_{\mathbb{R}}) \simeq \mathrm{Hom}(\mathbb{G}_{m\mathbb{C}}, T_{\mathbb{C}})$

$$\begin{array}{ccc} \cup & & \\ \mathrm{Hom}^0(\mathbb{S}, T_{\mathbb{R}}) & \longrightarrow & \mu_h \\ \rotatebox{90}{$\in$} & & \\ h & & \end{array}$$

in $X_*(R_{E/k})$, $w = -\sum \mu_\sigma$

$$w_h = h \circ w = -\mu_h - {}^c\mu_h, \quad \text{where } c \in \mathrm{Gal}(\mathbb{C}|\mathbb{R}) \text{ is the cplx conj.}$$

w_h is defined over $\mathbb{Q} \iff \gamma w_h = w_h$, $\forall \gamma \in \mathrm{Aut}(\mathbb{C}|\mathbb{Q})$

$$(\gamma - 1)(1+c)\mu_h = 0$$

Now let $\overline{R}_E = R_{E/\mathbb{Q}} / \ldots$ $\qquad E'|E$, $\begin{array}{ccc} R_{E'} & \xrightarrow{N_{E'/E}} & R_E \\ \downarrow & & \downarrow \\ \overline{R}_{E'} & \longrightarrow & \overline{R}_E \end{array}$

s.t. $X_*(\overline{R}_E) = X_*(R_{E|\mathbb{Q}}) / \{ (\gamma-1)(1+c)(\mu^E)_{\mathbb{C}} : \gamma \in \mathrm{Aut}(\mathbb{C}|\mathbb{Q}) \}$

Prop $T \mapsto \mathrm{Hom}^0(\mathbb{S}, T_{\mathbb{R}})$ is pro-corepresentable by

$$\left(\bar{R}_{\bar{\mathbb{Q}}} = \varprojlim \bar{R}_E, \ \mu^{\bar{E}} \right)$$

<u>Sketch of pf</u>

$$h : \mathbb{S} \longrightarrow T_{\mathbb{R}} \in \mathrm{Hom}^0(\mathbb{S}, T_{\mathbb{R}})$$

$$\Updownarrow$$

$$\mu_h : \mathbb{G}_{m\mathbb{C}} \longrightarrow T_{\mathbb{C}}, \quad (r-1)(1+c)\,\mu_h = 0, \quad \forall\, r \in \mathrm{Aut}(\mathbb{C}/\mathbb{Q})$$

Choose $E \subset \mathbb{C}$ # field $\quad \mu_h$ is defined

$$\mu'_h : \mathbb{G}_{m,E} \to T_E$$

$$\mu_h = (\mu'_h)_{\mathbb{C}} \quad \Updownarrow$$

$$\varphi : R_E \longrightarrow T$$

$$\mu'_h = \varphi_E \circ \mu^E$$

so $\quad \mu_h = \varphi_{\mathbb{C}} \cdot (\mu^E)_{\mathbb{C}}$

$$\begin{tikzcd}
(\mathbb{G}_m)_{\mathbb{C}} \arrow[r, "(\mu^E)_{\mathbb{C}}"] & (R_E)_{\mathbb{C}} \arrow[d, "\varphi_{\mathbb{C}}"] \\
(\bar{R}_E)_{\mathbb{C}} \arrow[ur] \arrow[r, "\bar{\varphi}"] & T_{\mathbb{C}}
\end{tikzcd}$$

$$\bar{\varphi} : \bar{R}_E \to T$$

Prop. $\bar{R}_{\bar{\mathbb{Q}}}$ is the tannakian gp of the cat. of rational Hodge str. w/ commutative MT gp.

$G_{MT}(V, h) = T$ comm is a torus $\quad \longleftrightarrow \quad h : \mathbb{S} \longrightarrow T_{\mathbb{R}}$ s.t. w_h is defined $/\mathbb{Q}$

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$\bar{R}_{\bar{\mathbb{Q}}} \longrightarrow T \curvearrowright (V, h)$ get a rep of $\bar{R}_{\bar{\mathbb{Q}}}$.

<u>Ex.</u> When $E|F$ CM ext'n,

$$\bar{R}_E = R_E / R_F^a, \quad \text{where} \quad R_F^a = \ker(Nm : R_F \longrightarrow \mathbb{G}_m)$$

$$\begin{tikzcd}
 & & R_E \arrow[r] & \bar{R}_E \\
1 \arrow[r] & R_F^a \arrow[r] \arrow[ur] & R_F \arrow[r] \arrow[u] & \mathbb{G}_m \arrow[r] & 1
\end{tikzcd}$$

If $E|\mathbb{Q}$ cubic but not totally real, $\bar{R}_E = R_E$.

<u>Lecture 6</u> k, $E|k$ finite sep. $R_{E|k} = \mathrm{Res}_{E|k}\, \mathbb{G}_m$

$$(\mathbb{G}_m)_E \xrightarrow{\mu^{E/k}} (R_{E/k})_E$$

(if $k = \mathbb{Q}$, we omit k from notation.

E-alg R

$$R^\times \longrightarrow (R \otimes_k E)^\times = R^\times \times (\dots)^\times$$

$$r \longmapsto (r, 1)$$

$k \hookrightarrow \Omega$ alg closed field

$$w : \mathbb{G}_m \longrightarrow R_{E/k}$$

$$R^\times \longrightarrow (E \otimes_k R)^\times$$

$$r \longmapsto 1 \otimes r^{-1}$$

$$X^*(R_{E/k})_\Omega = \mathrm{Hom}(\mathbb{G}_{m,\Omega}, (R_{E/k})_\Omega)$$

$\circlearrowleft$ $\mathrm{Aut}(\Omega/k)$

$$= \bigoplus_{\sigma : E \to \Omega} \mathbb{Z}[\sigma]$$

$$X_*(R_{E/k})_\Omega = \bigoplus_{\sigma : E \to \Omega} \mathbb{Z}\mu_\sigma$$

$E \xrightarrow{\sigma} \Omega$. e.g. $(\mu^{E/k})_\Omega = \mu_\sigma$

$f : \mathrm{Hom}(E, \Omega) \longrightarrow \mathbb{Z}$ $\qquad w_\Omega = -\sum \mu_\sigma$

$\mu_f := \sum f(\sigma)\mu_\sigma$

$E'|E|k$ $\qquad R_{E'/k} \xrightarrow{N_{E'/E}} R_{E/k}$

$$N_{E'/E} \circ \mu^{E'/k} = (\mu^{E/k})_{E'}$$

$\bar{k}/k$, $\quad \left(R_{\bar{k}/k} := \varprojlim_{E \subset \bar{k}} R_{E/k}, \ \mu^{\bar{k}/k} \right)$

pro-corepresents $\quad k$-tori $\longrightarrow$ abel. gp

$$T \longmapsto X_*(T)_{\bar{k}}$$

$$\mathrm{Hom}(R_{\bar{k}/k}, T) \longrightarrow X_*(T)_{\bar{k}}$$

$$\varphi \longmapsto \varphi \circ \mu^{\bar{k}/k}$$

$T = R_{F/k}$, $\quad \Phi \subset \mathrm{Hom}_k(F, \bar{k})$, $\quad \mu_\Phi := \mu_{\mathbb{1}_\Phi} = \sum_{\sigma \in \Phi} \mu_\sigma \in X_*(T)$

$$R_{\bar{k}/k} \longrightarrow R_{F/k}$$

reflex field of Φ

$$\left\{ \gamma \in \mathrm{Gal}(\bar{k}|k) : \gamma(\mu_\Phi) = \mu_\Phi \right\} \longleftrightarrow F_\Phi \subset \bar{k}$$

$\Leftrightarrow \gamma(\Phi) = \Phi$

$$\begin{tikzcd}
 & (R_{F_\Phi/k})_{F_\Phi} \arrow[r, "k\text{-rat'l}"] & (R_{F/k})_{F_\Phi} \\
\mathbb{G}_{m,F_\Phi} \arrow[ur, "\mu^{F_\Phi/k}"] \arrow[urr, bend right] & &
\end{tikzcd}$$

$$F_\Phi = \left\{ \sum_{a \in \Phi} \sigma(a) : a \in F \right\}$$

$k = \mathbb{Q}$, T $\mathbb{Q}$-torus

$$\underset{\substack{\| \\ \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m}}{\mathrm{Hom}^\circ(\mathbb{S}, T_{\mathbb{R}})} = \left\{ h \in \mathrm{Hom}(\mathbb{S}, T_{\mathbb{R}}) : \ w_h : (\mathbb{G}_m)_{\mathbb{R}} \xrightarrow{w} \mathbb{S} \xrightarrow{h} T_{\mathbb{R}} \right\}$$

is defined over $\mathbb{Q}$

$$\begin{array}{c} \| \\ \mathrm{Hom}((\mathbb{G}_m)_{\mathbb{C}}, T_{\mathbb{C}}) \end{array}$$

$$\mu_h \in \mathrm{Hom}((\mathbb{G}_m)_{\mathbb{C}}, T_{\mathbb{C}})$$

$$\| $$

$$\left[\mathbb{G}_{m,\mathbb{C}} \xrightarrow{\mu^{\mathbb{C}/\mathbb{R}}} \mathbb{S}_{\mathbb{C}} \xrightarrow{h_{\mathbb{C}}} T_{\mathbb{C}} \right]$$

$w_h = (1+c)\mu_h$, $c \in \mathrm{Aut}(\mathbb{C}/\mathbb{R}) \subset \mathrm{Aut}(\mathbb{C})$

w_h is defined $/\mathbb{Q}$ $\iff$ $(\gamma - 1)(1+c)\mu_h = 0$, $\forall\ \gamma \in \mathrm{Aut}(\mathbb{C})$

$\mathbb{Q} \subset \overline{\mathbb{Q}} \subset \mathbb{C}$, $\quad F_{\mu_h} = \overline{\mathbb{Q}}^{\{\gamma \in \mathrm{Aut}(\mathbb{C}) : \gamma\mu_h = \mu_h\}} \overset{\sigma_0}{\hookrightarrow} \mathbb{C}$

$$\begin{tikzcd} \mathbb{G}_m \arrow[r, "\mu_h"] & T_{\mathbb{C}} \\ \mu_{\sigma_0} = \mu^{F_{\mu_h}} \arrow[r, mapsto] & \mu_h \\ R_{F_{\mu_h}} \arrow[r, "\psi"] \arrow[dr] & T \\ & \overline{R}_{F_{\mu_h}} \arrow[u, "\psi"'] \end{tikzcd}$$

$\overline{R}_E \twoheadleftarrow R_E$

$X_*(\overline{R}_E) = X_*(R_E) \big/ (\gamma - 1)(c+1)\mu_{\sigma_0}$

$\overline{R}_{\overline{\mathbb{Q}}} = \varprojlim_{E \subset \mathbb{C}} \overline{R}_E$ under norm map

μ_{σ_0}

$\overline{\mathbb{Q}} \subset \mathbb{C}$

Prop (1) $(\overline{R}_{\overline{\mathbb{Q}}}, \mu_{\sigma_0})$ pro-corepresents $\quad$ $\mathbb{Q}$-~~tori~~ $\longrightarrow$ abel. gps

$T \longmapsto \mathrm{Hom}^\circ(\mathbb{S}, T_{\mathbb{R}})$

(2) $\mathrm{Rep}_{\mathbb{Q}}(\overline{R}_{\overline{\mathbb{Q}}}) \simeq$ $\mathbb{Q}$-pure Hodge str. w/ comm. MT gps

Eg. (1) E/F CM field, $F = E^{*=1}$, $\bar{R}_E = R_E / R_F^a$.

$$R_F^a = \ker\left(R_F \xrightarrow{Nm} \mathbb{G}_m\right)$$

(2) E cubic, not totally real, $\bar{R}_E = R_E$.

Def. Let E be a no. field. The Serre torus of E is the quotient

$$\begin{tikzcd} R_E \arrow[r, two heads] \arrow[dr, two heads] & \mathcal{T}_E \\ & \bar{R}_E \arrow[u, two heads] \end{tikzcd}$$

s.t. $X_*(\mathcal{T}_E)_{\mathbb{C}} = X_*(R_E)_{\mathbb{C}} \Big/ \left(\begin{matrix} (\gamma-1)(c+1)(\mu^E)_{\mathbb{C}} \\ (c+1)(\gamma-1)(\mu^E)_{\mathbb{C}} \end{matrix} \right)$

$\gamma \in \mathrm{Aut}(\mathbb{C})$

$E' | E$,

$$\begin{tikzcd} R_{E'} \arrow[r, "N_{E'/E}"] \arrow[d] & R_E \arrow[d] \\ \mathcal{T}_{E'} \arrow[r] & \mathcal{T}_E \end{tikzcd}$$

$\left(\mathcal{T} := \varprojlim_{E \subset \mathbb{Q}^{alg}} \mathcal{T}_E, \mu \right)$

Prop $(\mathcal{T}_E, \mu^E)$ μ^E is defined over the max'l CM subfield of $E^c \subset E$,

$$\mathcal{T}_E = \mathcal{T}_{E^c} = \bar{R}_{E^c}.$$

In general, $\bar{R}_E \underset{\neq}{\twoheadrightarrow} \mathcal{T}_E$.

$(\mathcal{T}, \mu^{\mathbb{Q}^{CM}})$ $\mathbb{Q}^{CM} \subset \mathbb{Q}^{alg}$

max. CM subfield of $\mathbb{Q}^{alg}$

$\mathbb{Q}^{alg} = \bar{\mathbb{Q}} \hookrightarrow \mathbb{C}$

Thm $\mathrm{Rep}_{\mathbb{Q}}(\mathfrak{G}) \simeq \mathrm{CMHS}_{\mathbb{Q}}$ ← polarizable pure $\mathbb{Q}$-HS w/ comm. MT gp

Sketch pf. Recall if $G = G_{MT}(V,h)$ is associated to (V,h) polarizable

then $\mathrm{ad}\, h(i)$ is a Cartan involution $\left(G/w_h(\mathbb{G}_m)\right)_{\mathbb{R}}$

i.e. $\{ g \in \left(G/w_h(\mathbb{G}_m)\right)_{\mathbb{C}} : h(i)\, \bar{g}\, h(i)^{-1} = g \}$ is compact.

Now $G = T$. is a torus.

Claim $\gamma c = c\gamma : X_*(T_{\mathbb{C}})^{\mu_h} \longrightarrow X_*(T_{\mathbb{C}})$, $\gamma \in \mathrm{Aut}(\mathbb{C})$

$c \in \mathrm{Aut}(\mathbb{C}/\mathbb{R})$, $c^2 = 1$

$\left(T/w_h(\mathbb{G}_m)\right)_{\mathbb{R}}$ is compact.

$$\left(X_*(T_{\mathbb{C}})\right)^{c=1}_{\mathbb{Q}} = \mathbb{Q}\cdot w_h$$

$$\Downarrow \qquad \Uparrow \quad \| $$

$$X_*(T_{\mathbb{C}})^{\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})}_{\mathbb{Q}}$$

$1+c : X_*(T_{\mathbb{C}}) \otimes \mathbb{Q} \longrightarrow \mathbb{Q}\cdot w_h \subset X_*(T_{\mathbb{C}}) \otimes \mathbb{Q}$

$\Updownarrow$

$\chi \in X^*(T_{\mathbb{C}})^{c=1}_{\mathbb{Q}} = X^*(T_{\mathbb{C}})^{\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})}_{\mathbb{Q}}$

$\gamma \circ \chi = \chi$, $\forall \gamma$ $\iff$ $\gamma(1+c) = (c+1)\gamma$

We know $\gamma(c+1)\mu_h = (c+1)\mu_h$, $\forall \gamma$ $\Rightarrow$ $(c+1)\gamma\mu_h = (c+1)\mu_h$

$(c+1)(\gamma - 1)\mu_h = 0$

$$\begin{tikzcd} \bar{R}_{\mathbb{Q}} \arrow[rr] \arrow[dr] & & T \\ & \mathfrak{G} \arrow[ur] & \end{tikzcd}$$

If $(V, h) \in CMHS_{\mathbb{Q}}$,

$$\mathfrak{G} \twoheadrightarrow T = G_{MT}(V, h)$$

Recall the example, E CM field, Φ CM type

$$\begin{array}{ccc} E & & \Phi \\ \cup & & \cap \\ F = E^{*=1} & & \mathrm{Hom}(E, \mathbb{C}) \end{array}$$

$$(E, h_\Phi)$$

$$\mu_\Phi = \sum_{\sigma \in \Phi} \mu_\sigma \in X_*((R_E)_{\mathbb{C}})$$

$$\begin{tikzcd} & G_{MT}(E, h_\Phi) \arrow[r, phantom, "\subset"] & GU_{E/F}(1) \arrow[r, phantom, "\subset"] & R_E \\ R_{E_\Phi}/R^a_{F_\Phi} = \mathfrak{G}_{E_\Phi} \arrow[u] & & \{t \in R_E : N_{E/F}(t) \in \mathbb{Q}\} \arrow[u, phantom, "\shortparallel"] & \\ E_\Phi \subset \mathbb{C} & R_{E_\Phi} \arrow[u] \arrow[urr] & & \end{tikzcd}$$

<u>Prop</u> $CMHS_{\mathbb{Q}}$ is generated by $\{(E, h_\Phi)\}_{(E,\Phi)}$ under $\oplus$, $\otimes$, duals, subquotients.

e.g.

(E, h_{Φ_i}) $\quad \bigotimes_i (E, h_{\Phi_i})^{\varepsilon_i}$, $\varepsilon_i \in \{\pm 1\}$ $\quad \bigotimes_A E$, $\quad A = \bigotimes_i E$

$$(E, \mu_h = \sum \varepsilon_i \mu_{\Phi_i})$$

<u>Sketch of proof</u> $(V, h) \in CMHS_{\mathbb{Q}}$ irreducible, $\mathfrak{G} \longrightarrow GL(V)$

$\mathrm{End}(V,h)$ is commutative

$\|$

E

$\Rightarrow E$ is a CM field, $[E:\mathbb{Q}] = \dim V$

$$\mathbb{G} \longrightarrow G_{MT}(V,h) \subset GU_{E/F}(1) \subset R_E$$

$\mu_h = \sum a_\sigma \mu_\sigma$, $\quad a_\sigma + a_{c\sigma} = -m$, $\quad m$ wt of (V,h)

$= \sum \varepsilon_i \mu_{\Phi_i}$ $\quad c(\mu_h) + \mu_h = -w_h$

<u>Lecture 7</u>. $HS_\Lambda \subset \mathrm{Fil}_{\mathbb{C}/\Lambda} = \{(V, F^\bullet) : \cdots\}$

<u>Lemma</u> Fix V finite free Λ-mod.

Consider $\mathrm{Fil}_V : \mathrm{CAlg}_\Lambda \longrightarrow \mathrm{Sets}$

$R \longmapsto \mathrm{Fil}_{R/\Lambda}$

$\mathrm{Fil}_{R/\Lambda} = \{F^\bullet(V\otimes R) : F^i \supset F^{i+1}, F^i/F^{i+1}$ finite proj. R-mod

$F^i = V\otimes R, i \ll 0, F^i = 0, i \gg 0\}$

is represented by a disjoint union of projective schemes $/\Lambda$

$\mathrm{Fil}_V = \coprod_{(d_i)} \mathrm{Fil}_{V,(d_i)}$, $\quad \mathrm{Fil}_{V,(d_i)} = \{F^\bullet : \mathrm{rk}(F^i/F^{i+1}) = d_i\}$

$\uparrow$

partial flag variety

Ex. $V = F^0 \underset{1}{\supset} F^1 \underset{d-1}{\supset} F^2 = 0$

$d_0 = 1, d_1 = d-1, d_i = 0, i \neq 0, 1$, $\quad \mathrm{Fil}_{V,(d_i)} = \mathbb{P}(V^*)$

Rmk. (1) $\exists$ natural bijection

$$\left\{ (d_i)_{i\in\mathbb{Z}} : \begin{array}{l} d_i \in \mathbb{Z}_{\geq 0} \\ \sum d_i = d \end{array} \right\} \simeq \left\{ \underset{\mu}{\underset{\shortparallel}{(m_1, \cdots, m_d)}} \in \mathbb{Z}^d : e \geq m_1 \geq m_2 \geq \cdots \geq m_d \right\} \simeq X_*(T)^*$$

$$\mu \longrightarrow \mathbb{G}_m \to GL_d, \quad t \mapsto \begin{pmatrix} t^{m_1} & & \\ & \ddots & \\ & & t^{m_d} \end{pmatrix}$$

$$(d_i) \longmapsto (\underbrace{-i, \cdots, -i}_{d_i}, \underbrace{-i-1, \cdots, -i-1}_{d_{i+1}}, \cdots)$$

(2) If $(V, F^\cdot)$ is a Hodge str. of wt m, $d_i = \dim_{\mathbb{C}} V^{i,m-i}_{\mathbb{C}} = h^{i,m-i}$

Rmk Rees construction

$$\left(F^\cdot \text{ of } V \underset{\Lambda}{\otimes} R \right) \longrightarrow \left\{ (\mathcal{V}, \mathcal{V}', \beta) : \begin{array}{l} \mathcal{V}, \mathcal{V}' \text{ finite proj. } R[t]\text{-mods}, \\ \beta : \mathcal{V} \underset{R[t]}{\otimes} R[t, t^{-1}] \xrightarrow{\sim} \mathcal{V}' \underset{R[t]}{\otimes} R[t, t^{-1}] \end{array} \right\}$$

$$V \longmapsto \mathcal{V} = V \underset{\Lambda}{\otimes} R[t]$$

$$\mathcal{V}' = \sum_i F^i V_R \otimes t^{-i} R[t] \subset V \underset{\Lambda}{\otimes} R[t, t^{-1}]$$

Fix $(V, \operatorname{rk}_\Lambda V = d, (d_i))$

$$\tilde{D}_{V,(d_i)} = \{ x \in \mathrm{Fil}_{V,(d_i)}(\mathbb{C}) : (V, F_x^\cdot) \text{ is a pure Hodge str.} \}$$

Lemma (1) $\tilde{D}_{V,(d_i)}$ is open in $\mathrm{Fil}_{V,(d_i)}(\mathbb{C})$

pre-period domain

$\mathbb{C}$ equipped w/ analytic top.

(2) Regard $\mathrm{Fil}_{V,(d_i)/\mathbb{C}}$ as an $\mathbb{R}$-scheme (via Weil restriction), then $\tilde{D}_{V,(d_i)}$ is an open subscheme.

$\boxed{\text{Suppose } d_i = d_{m-i}}$

$$\begin{tikzcd}
\tilde{D}_{V,(d_i)} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{U} = \{ (F_1^\cdot, F_2^\cdot) \in \mathrm{Fil}_{V,(d_i)} \times \mathrm{Fil}_{V,(d_i)} : F_1^p \cap F_2^{m+1-p} = (0), \forall p \} \arrow[d] \\
\mathrm{Fil}_{V,(d_i)} \arrow[r] & \mathrm{Fil}_{V,(d_i)} \times \mathrm{Fil}_{V,(d_i)} \\
F^i \arrow[r, mapsto] & F^i, \ \overline{F^{m+1-i}}
\end{tikzcd}$$

Ex. $V = \Lambda^2$, $d_0 = 1$, $d_1 = 1$ $\qquad V_{\mathbb{C}} = F^0 \overset{1}{\supset} F^1 \overset{1}{\supset} F^2 = 0$

$h^{0,1} = h^{1,0} = 1$

$$D_{V,(d_i)} \subset \mathrm{Fil}_{V,(d_i)} = \mathbb{P}^1$$
$$\| $$
$$\mathcal{H}^{\pm 1} = \mathbb{P}^1(\mathbb{C}) - \mathbb{P}^1(\mathbb{R})$$

Ex. $V = \Lambda^2$, $d_0 = 1$, $d_2 = 1$ $\qquad V_{\mathbb{C}} = F^0 \overset{1}{\supset} F^1 = F^2 \overset{1}{\supset} F^3 = F^4 \cdots$

$h^{0,2} = h^{2,0} = 1$

$$D_{V,(d_i)} \subset \mathrm{Fil}_{V,(d_i)} = \mathbb{P}^1$$
$$\| $$
$$\mathbb{P}^1(\mathbb{C}) - \mathbb{P}^1(\mathbb{R})$$

$V, (d_i), \eta : V \times V \to \Lambda$ bilinear

$$\underset{\substack{\uparrow \\ \text{period} \\ \text{domain}}}{D_{V,(d_i),\eta}} = \{ F^\cdot : (V, F^\cdot) \text{ is a pure HS of wt } m, \ \pm (2\pi i)^{-m} \eta =: Q : V \times V \to \Lambda(m) \text{ is a polarization} \}$$

$$\subset \tilde{D}_{V,(d_i)} \subset \mathrm{Fil}_{V,(d_i)}$$

Rmk / Lemma If $(V, F^\bullet, Q)$ is a polarized Hodge str. of wt w,

$$Q_{\mathbb{C}}(F^p, F^{m+1-p}) = 0 \quad \Rightarrow \quad F^{m+1-p} = (F^p)^\perp.$$

$$F^p = \bigoplus_{p' \geq p} V^{p', m-p'}, \qquad Q_{\mathbb{C}} : V^{p,q} \times V^{p',q'} \longrightarrow \mathbb{C}$$

$$\overset{\shortparallel}{0} \quad \text{unless} \quad \begin{cases} p' = q \\ q' = p \end{cases}$$

$$[\, H_{\mathbb{C}}(v, w) = Q_{\mathbb{C}}(v, h(i) w) \,]$$

$$\begin{array}{ccc} D_{V,(d_i),\eta} & \subset & \widetilde{D}_{V,(d_i)} \\ \text{open in analytic top.} \cap & & \cap \\ \mathrm{Fil}_{V,(d_i),\eta}(\mathbb{C}) & \subset & \mathrm{Fil}_{V,(d_i)}(\mathbb{C}) \end{array}$$

$$\{F^\bullet : \eta_{\mathbb{C}}(F^p, F^{m+1-p}) = 0\}$$

$$F^p \cap \overline{F}^{m+1-p} = 0 \quad \leftarrow \text{open}$$

$$Q_{\mathbb{C}} : V^{p,q} \times V^{p',q'} \longrightarrow \mathbb{C} \quad = 0 \quad \text{unless} \quad p' = q,\ q' = p$$

$H_{\mathbb{C}}$ = non-deg. Hermitian form, signature constant on each comp.

Ex $h^{1,0} = h^{0,1} = g$

$\mathrm{rk}\, V = 2g$

$$\eta : V \times V \longrightarrow \Lambda \qquad F^0 \overset{g}{\supset} F^1 \overset{g}{\supset} F^2 = 0$$

$$V^*$$

$\alpha_1, \dots, \alpha_g,$
$\beta_1, \dots, \beta_g$ $\quad \Lambda$ - basis of V

$$\eta(\alpha_i, \alpha_j) = \eta(\beta_i, \beta_j) = 0$$

$$\eta(\alpha_i, \beta_j) = \delta_{ij}$$

Choose a basis $\omega_1, \dots, \omega_g$ of F^1

$$\left\{ \begin{matrix} g\text{-dim'l subspaces} \\ \text{of } V^* \end{matrix} \right\} \longleftrightarrow \{ A = (\alpha_i, w_j),\ B = (\beta_i, w_j) \} / GL_g$$

$$(A \ \ B) \begin{pmatrix} & -I \\ I & \end{pmatrix} \begin{pmatrix} A^T \\ B^T \end{pmatrix} = 0,$$

$$i (A \ \ B) \begin{pmatrix} & -I \\ I & \end{pmatrix} \begin{pmatrix} \bar{A}^t \\ \bar{B}^t \end{pmatrix} \underset{\text{or } < 0}{> 0} \leftarrow \text{pos. def or neg. def}$$

<u>Claim</u> A invertible. $\{A, B\} / GL_g \simeq \{I, Z\}$

① $Z = Z^t$ (Z symmetric)

② $i(Z - Z^t) \underset{\text{or } < 0}{> 0} \iff \mathrm{Im}(Z)$ pos/neg definite.

$$\mathcal{H}_g^{\pm 1} = \left\{ (g \times g)\text{-cplx sym. mat. } Z : \mathrm{Im}(Z) \underset{\text{or } < 0}{> 0} \right\}$$

$g = 1$, $\mathcal{H}^{\pm 1}$.

<u>Ex</u>. $h^{-2,0} = a$, $h^{-1,-1} = b$, $h^{0,-2} = a$, $m = 2$

$d = 2a + b$, $\eta : V \times V \to \Lambda$ non-deg. symmetric.

$$V_{\mathbb{C}}^* = F^0 \underset{a}{\supset} F^1 \underset{b}{\supset} F^2 \underset{a}{\supset} F^3 = 0.$$

$$V_{\mathbb{R}}^* = \underbrace{V_{\mathbb{C}}^{2,0} \oplus V_{\mathbb{C}}^{0,2}}_{V_0} \oplus \underbrace{V_{\mathbb{C}}^{1,1}}_{V_1}$$

$\{ F : F^2 \subset (F^2)^\perp \}$

If $\pm (2\pi i)^{-2} \eta$ is a polarization.

$\eta_{\mathbb{R}} |_{V_0} < 0$, $\eta_{\mathbb{R}} |_{V_1} > 0$

or $\eta_{\mathbb{R}} |_{V_0} > 0$, $\eta_{\mathbb{R}} |_{V_1} < 0$. $\eta_{\mathbb{R}}$ signature $(2a, b)$

$D_{V,(2a,b),\eta}$.

Assume $a=1$. Choose a basis of $V_{\mathbb{R}}$, $f_1, f_2, e_1, \dots, e_b$

$\eta(f_1, f_1) = \eta(f_2, f_2) = 0$, $\quad \eta(f_1, f_2) = 1$

$\eta(e_1, e_1) < 0$ $\qquad \eta(f_i, e_j) = 0$

$\eta(e_i, e_i) > 0$, $i > 1$ $\qquad \eta(e_i, e_j) = 0$

$$F^2 \longleftrightarrow \{ (1, z, w_1, \dots, w_b) : z, w_i \in \mathbb{C}, \ 2z = w_1^2 - w_2^2 - \dots - w_b^2 \}$$

$$z + \bar{z}^t = |w_1|^2 - |w_2|^2 - \dots - |w_n|^2 \ \begin{matrix} > 0 \\ < 0 \end{matrix}$$

$$\subset \{ \mathbb{C}^b : (w_1 - \overline{w_1})^2 - (w_2 - \overline{w_2})^2 - \dots - (w_n - \overline{w_n})^2 \ \begin{matrix} > 0 \\ < 0 \end{matrix} \}$$

<u>Lecture 8</u>. Let G = conn'd reductive group $/ \Lambda = \mathbb{Q}$

$\mathrm{Fil}_{R/\Lambda}(G)$

$\{ V \in \mathrm{Rep}_\Lambda(G)$, $F^\cdot$ of $V \otimes R$ compatible w/ tensor prod, direct sum $\}$

$$F^i(V \otimes W) = \sum_j F^j(V) \otimes_R F^{i-j}(W)$$

$$F^i(V_R \oplus W_R) = F^i(V_R) \oplus F^i(W_R)$$

<u>Prop.</u> $\mathrm{Fil}_G : \mathrm{CAlg}_\Lambda \longrightarrow \mathrm{Sets}$

$R \longrightarrow \mathrm{Fil}_{R/\Lambda}(G)$ is represented by a disjoint union of proj. schemes $/\Lambda$

If $k|\Lambda$ a finite ext'n s.t. G_k is split, then $\mathrm{Fil}_G \otimes_\Lambda k = \coprod_{\mu \in \mathrm{Hom}(\mathbb{G}_m, G)/G} \mathrm{Fil}_{G,\mu}$

If we pick $\mu : \mathbb{G}_m \to G$, $P_\mu = \{ g \in G : \lim_{t \to 0} \mu(t)\, g\, \mu(t)^{-1} \text{ exists} \} \subset G$

is a parabolic subgp. $\mathrm{Fil}_{G,\mu} = G/P_\mu$.

One way to prove it:

$$\mathrm{Rep}_\Lambda(G) \xrightarrow[\otimes\text{-functor}]{\text{exact}} \mathrm{Fil}_{R/\Lambda}$$

$$\downarrow$$

$$\{ (V, V', \beta) \}$$

$$//$$

$$R[t]$$

$$\mathrm{Fil}_G = \mathrm{Gr}_G^{\mathbb{G}_m}$$

Now suppose $G \subset GL(V)$

$$\left(\mathrm{Rep}_\Lambda(G) = \langle V \rangle^{\otimes}, \quad (\Lambda \text{ field}) \right)$$

$$G = \{ g \in GL(V) : g(t) = t, \quad \forall t \in T^{\cdot\cdot}(V)^G \}$$

$$\overset{\shortparallel}{T}$$

<u>Lemma</u> Let $F^\bullet \in \mathrm{Fil}_{R/\Lambda}(V)$. Then TFAE:

(1) $G_R \wedge \mathrm{Aut}(V_R, F^\bullet) =: P$ is a parabolic subgroup of G_R

Let $U \subset P$ be the unipotent radical, $U = \{ g \in G_R : g \equiv 1 \text{ on } \mathrm{gr}_F(V) \}$

$$\begin{tikzcd} P/U \arrow[rr] & & GL(\mathrm{gr}_F V_R) \\ & \mathbb{G}_m \arrow[ul] \arrow[ur] & \end{tikzcd}$$

(2) $\exists\ \mathbb{G}_m \to G_R \subset GL(V_R)$ $\quad (\Leftrightarrow V_R = \bigoplus V_R^i)$

s.t. $F^i V_R = \bigoplus_{i' \geq i} V_R^{i'}$

(3) $V_R = \bigoplus_i V_R^i$ s.t. $F^i V_R = \bigoplus_{i' \geq i} V_R^{i'}$ & $T \subset T^{\circ,\circ}(V)^\circ$

<u>Def</u> Such F is called a <u>G-splitable filtration</u> on V_R.

<u>Pf</u> (2) $\Leftrightarrow$ (3):

$$\begin{tikzcd} \mathbb{G}_m \arrow[rr] \arrow[dr] & & GL(V_R) \\ & G_R \arrow[ur] & \end{tikzcd}$$

(1) $\mu : \mathbb{G}_m \to G$

$P_\mu = \{ g \in G :$

$\overset{\shortparallel}{P}$

$G \subset GL(V)$

$\mathrm{Fil}'_G : \mathrm{CAlg}_\Lambda \to \mathrm{Set}$

$R \longmapsto G$-splitable filtrations on V_R

<u>Prop.</u> $\mathrm{Fil}_G \simeq \mathrm{Fil}'_G \subset \mathrm{Fil}_V$

<u>Rmk.</u> (1) Let F' be a G-splittable filtration on V_R. Let μ_1, μ_2 be two such cochars. Then étale locally on R, μ_1, μ_2 are conjugate.

given $\mu \in \mathrm{Hom}(\mathbb{G}_m, G)/G$

G-splittable μ-filtration

$\mathrm{Fil}_{G,\mu}$

$(\Lambda = \text{field})$ $\Lambda \subset \Omega$ $\quad \mathrm{Hom}(\mathbb{G}_{m,\Omega}, G_\Omega)/G_\Omega$

$\mathrm{Fil}_{G,\mu}$ is defined over $E_\mu \subset \Omega$

field of def'n of the conj. class of μ

called the reflex field.

(2) $\mathbb{G}_m \xrightarrow{\mu} G \longrightarrow GL(V)$ $\quad d = \mathrm{rk}(V)$

$(m_1 \geq \cdots \geq m_d) \Leftrightarrow (d_i)$

$\mathrm{Fil}_{G,\mu} \subset \mathrm{Fil}_{V,(d_i)} \otimes E_\mu$

There could be different μ giving same (d_i)

$\mathrm{Fil}_{G,\mu} \simeq \mathrm{Fil}_{G,n\mu}$.

(3) if $G = \mathrm{Aut}(V, \eta)$,

$\mathrm{Fil}_{G,\mu}$ parametrized - isotropic filtration.

<u>Lemma</u>. Let $x \in \mathrm{Fil}_{G,\mu}(\mathbb{C})$, TFAE.

(1) for every $V \in \mathrm{Rep}_\Lambda^{fr}(G)$, (V, F_x) is in HS_Λ

(2) $\exists$ a faithful rep'n V of G s.t. (V, F_x) in HS_Λ.

Then we can define: pre-Mumford-Tate's domain

$\widetilde{D}_{G,\mu} = \{x \in \mathrm{Fil}_{G,\mu}(\mathbb{C})$ satisfying equiv. conditions$\}$

$= \mathrm{Fil}_{G,\mu}(\mathbb{C})^{an} \cap \widetilde{D}_{V,(d_i)}$ in $\mathrm{Fil}_{V,(d_i)}(\mathbb{C})$

$\Rightarrow \widetilde{D}_{G,\mu}$ is open in $\mathrm{Fil}^{an}_{G,\mu}$, real subscheme

$\widetilde{D}_{G,\mu} \ni x$, $\quad h_x : \mathbb{S} \to GL(V_{\mathbb{R}})$ up to conj.

$\mu : \mathbb{G}_{m,\mathbb{C}} \to GL(V_{\mathbb{C}})$

μ_{h_x} is conj. to μ

<u>Lemma</u> $h_x : \mathbb{S} \to \mathbb{G}_m$ and μ_{h_x} is conj. to μ in $G_{\mathbb{C}}$.

$G_{MT}(V, h_x)$

$\cap$

G

$(V, F^\bullet)$, $F^\bullet T^{0,0}(V)$

$T = T^{\bullet,\bullet}(V)^G$

$\mu : \mathbb{G}_m \to GL(V)$

i.e. $T_{\mathbb{C}} \subset T^{\bullet,\bullet}(V)$ $\quad \mu(\mathbb{G}_m)$ deg 0

$\cap$

$F^0 T^{\bullet,\bullet}(V)$

$T_{\mathbb{R}} \subset F^0 T^{\bullet,\bullet}(V) \cap F_x T^{\bullet,\bullet}(V)$

$\mu_{h_x} : \mathbb{G}_m \to G_{\mathbb{C}}$ inducing filtration which is just F_x.

$\widetilde{D}_{G,\mu} = \{ h_x : \mathbb{S} \to G_{\mathbb{R}} : \mu_{h_x} \sim \mu \text{ in } G(\mathbb{C}) \}$

Lemma $\Rightarrow$ $\widetilde{D}_{G,\mu} \subset \{ h_x : \mathbb{S} \to \mathbb{G}_m \dots \}$

$\mathbb{S} \to G_{\mathbb{R}} \subset GL(V_{\mathbb{R}}) \subset \widetilde{D}_{V,(d_i)}$

(V, h) $\quad \cap \mathrm{Fil}_{G,\mu}$

<u>Ex.</u> $G = T / \Lambda \subset \mathbb{R}$, $\quad \mu : \mathbb{G}_m \to G_{\mathbb{C}}$. $\widetilde{D}_{G,\mu} = \{*\}$

Consider $\mathrm{Ad}: G \to GL(\mathfrak{g})$

Let $\mathrm{Kil}: \mathfrak{g} \otimes \mathfrak{g} \to \mathbb{A}$, $(x,y) = \mathrm{tr}(\mathrm{ad}\, x\ \mathrm{ad}\, y)$

be the Killing form on $\mathfrak{g}$.

<u>Lemma</u> Let $x \in \widetilde{D}_{G,P}$. TFAE.

(1) $(\mathfrak{g}, \mathrm{Ad} \circ h_x, -\mathrm{Kil})$ is a polarized $\mathbb{R}$-Hodge str. of wt 0.

(2) $w_{h_x}(\mathbb{G}_m) \subset (Z_G)_{\mathbb{R}}$ and $\mathrm{Ad}(h(i))$ is a Cartan involution of $(G_{ad})_{\mathbb{R}}$

(3) For every rep'n $G \to GL(V)$ w/ a central char. $\exists$ a character

$\chi_{\mathbb{R}}: G_{\mathbb{R}} \to \mathbb{G}_m$ & a G-inv bilinear form

$\psi: V_{\mathbb{R}} \times V_{\mathbb{R}} \to \chi_{\mathbb{R}}$ s.t. ψ is a polarization of $(V_{\mathbb{R}}, \rho \circ h_x)$.

<u>Lecture 9</u>. $G/_{\mathbb{R}} \hookrightarrow GL(V)$ connected reductive group, $\{\mu: \mathbb{G}_m \to G_{\mathbb{C}}\}/G(\mathbb{C})$

$$D_{G,\mu} \subset \widetilde{D}_{G,\mu} \subset Fl_{G,\mu} \simeq G/P_\mu$$

$$\widetilde{D}_{G,\mu} = \{x : (V, F_x^\bullet) \text{ is a HS}\} \simeq \{h: \mathbb{S} \to G,\ \mu_h \sim \mu\}$$

$$x \mapsto (V, F_x^\bullet)$$

G-splittable filtration

<u>Prop</u>. Fix $x \in \widetilde{D}_{G,\mu}$, TFAE:

(1) $(\mathfrak{g}_{ad}, -\mathrm{Kil}, \mathrm{Ad} \circ h_x)$ is a polarized pure $\mathbb{R}$-HS of wt 0

(2) $w_{h_x}(\mathbb{G}_m) \subset Z_G$, $\mathrm{Ad}\, h(i)$ is a Cartan inv'n of G_{ad}

(3) For every rep'n (W,ρ) of G w/ a central char. $\exists$ a $\chi : G \to \mathbb{G}_m$

& $B : W \otimes W \to \chi$ G-equivariant. s.t.

$(W, \pm(2\pi i)^m B, \rho \circ h)$ is a polarized HS.

(4) Replace "every" in (3) by one <u>almost faithful</u> rep. of G

$\uparrow$

$\ker(G \to GL(W)) \subset Z_G$

Pf. (1) $\Rightarrow$ (2) done before.

(2) $\Rightarrow$ (3) First assume W is irred.

$\exists$ W_0 irred rep'n of $G_{\mathbb{C}}$

s.t. $W_{\mathbb{C}} = W_0$ or $W_0^{\oplus 2}$ or $W_0^{\oplus 4}$

$(G_h)_{der} \curvearrowright W_0$, $\exists!$ up to $\mathbb{R}_{>0}$-multiple

positive def. $(G_h)_{der}$-inv. hermitian form

H_0 on W_0.

$H_0(gv, gw) = H_0(v,w)$, $\forall g \in (G_h)_{der}$

$h = h_x$

$Z_G \subset G_h = \{ g \in G(\mathbb{C}) : h(i)\bar{g}h(i)^{-1} = g \}$

$h(S) \subset$

Ad $h(i)$ is a Cartan-involution of G_{ad}

if $(G_h)/Z_G$ is compact.

$(G_h)_{der}$ compact

$\Rightarrow$ $\exists$ $\chi : G_h \to \mathbb{G}_m$ s.t. $H_0(gv, gw) = \chi(g) H_0(v,w)$

(3) $\Rightarrow$ (1) ✓

(3) $\Rightarrow$ (4) ✓

(4) $\Rightarrow$ (2) ✓

Similar to (1) $\Rightarrow$ (2)

$\chi : G_h \to \mathbb{G}_m$

$z \in Z_G$, $c_\rho = \rho|_{Z_G} : Z_G \to \mathbb{G}_m$

$z \in Z_G$

$c_\rho(z)^2 H(v,w) = H(\rho(z)v, \rho(z)w) = \chi(z) H(v,w)$

$\chi|_{Z_G} = c_\rho^2$

$\mathbb{G}_m \xrightarrow{w_h} Z_G \xrightarrow{c_p} \mathbb{G}_m$, $r \mapsto r^m$ $\quad (W, \rho \circ h)$ is a pure HS of wt m.

In addition, let $G(\mathbb{C}) \longrightarrow G(\mathbb{C})$, $g \mapsto \underset{h(i)\bar{g}h(i)^{-1}}{\overset{\shortparallel}{g^c}}$ be the complex conj. of $G(\mathbb{C})$ w.r.t. G_h.

$$H(gv, g^c w) = \chi(g) H(v, w) \quad , g \in G(\mathbb{C})$$

Now def. $B : W \times W \longrightarrow \mathbb{R}$, $B(v, w) = \mathrm{Re}\, H(v, h(i)^{-1} w)$

① B is $(-1)^m$-sym. , $(v, w) \mapsto B(v, h(i) w)$ is sym.

② $(v, w) \mapsto B(v, h(i) w)$ is positive.

③ $B : W \otimes W \longrightarrow \mathbb{X}$ is G-equivariant

$$B(w, h(i) v) = \mathrm{Re}\, H(w, v) = \mathrm{Re}\, \overline{H(v, w)} = \mathrm{Re}\, H(v, w) = B(v, h(i) w)$$

$h(i)^2 = (-1)^m$

$$B(gv, gw) = \mathrm{Re}\, H(gv, h(i)^{-1} g w), \quad g \in G \quad , \bar{g} = g$$

$$= \mathrm{Re}\, H(gv, \underbrace{h(i)^{-1} \bar{g} h(i)}_{g^c} h(i)^{-1} w)$$

$$= \mathrm{Re}\left(\chi(g) H(v, \overset{h(i)^{-1}}{w})\right) = \chi(g) B(v, w)$$

$D_{G,\mu} = \{ x \in \widetilde{D}_{G,\mu} : h_x$ satisfying the equiv. conditions in the previous prop $\}$

$$\simeq \left\{ h : \mathbb{S} \longrightarrow G : \begin{array}{l} - \ w_h(\mathbb{G}_m) \subset Z_G \\ - \ \mu_h \sim \mu \\ - \ \mathrm{Ad}\, h(i) \text{ is a Cartan involut'n of } G_{ad} \end{array} \right\}$$

Def. Let S be a cplx ~~mfd~~ analytic space (or scheme). A filtered v.b. on S is $(V, F^\bullet)$ where

- V is a v.b. on S
- $F^i V \supset F^{i+1} V \supset \cdots$ filtration by subbundles s.t. $F^i V / F^{i+1} V$ is a v.b.

  $F^i V = V$, $i \ll 0$, $F^i V = 0$, $i \gg 0$.

Def Let S be a cplx mfd. A variation of Λ-HS consists of $(\mathbb{L}, V, \nabla, F^\bullet, \varepsilon)$ where

- $\mathbb{L}$ locally constant sheaf of finite free Λ-mod
- $(V, \nabla : V \to \Omega^1_S \otimes V)$ is a v.b. w/ a flat conn.
- $(V, F^\bullet)$ is a filtered v.b.

satisfying Griffith transversality $\nabla(F^i) \subset \Omega^1_S \otimes F^{i-1}$

- $\varepsilon : (\mathbb{L} \otimes \mathcal{O}_S, \nabla = 1 \otimes d) \simeq (V, \nabla)$

s.t. at each pt $s \in S$, $(\mathbb{L}_s, F^\bullet_s)$ is a pure Λ-HS.

A polarization of var. of Λ-HS of weight m is a morphism

$$Q : \mathbb{L} \otimes \mathbb{L} \to \Lambda(-m)$$

s.t. $\forall s \in S$, $Q_s : \mathbb{L}_s \otimes \mathbb{L}_s \to \Lambda(-m)$ is a polarization.

Ex. $\begin{array}{c} X \\ f \downarrow \\ S \end{array}$ f smooth proj. morphism of smooth var. $/\mathbb{C}$

$\mathbb{L} = R^m f_* \Lambda$, $V = R^m f_{dR,*} \mathcal{O}_X = (\mathcal{O}_X \xrightarrow{d} \Omega^1_{X/S} \xrightarrow{d} \overbrace{\Omega^2_{X/S}}^{F^2} \to \cdots)$

$(\mathbb{L} \otimes \mathcal{O}_S, d) \simeq (V, \nabla^{GM})$, $\nabla = \nabla^{GM} \leftarrow$ Gauß-Manin connection

$$F^\bullet V = Rf_*\left(0 \to \Omega^p_{X/S} \to \Omega^{p+1}_{X/S} \to \cdots\right)$$

Consider $\widetilde{D}_{G,\mu} \subset \mathrm{Fil}_{G,\mu} =: S \simeq G/P_\mu$

Let $V \in \mathrm{Rep}(G)$

Constant sheaf $\underline{V}$, $\left(\mathcal{V} = V \otimes \mathcal{O}_{\widetilde{D}_{G,\mu}},\ \nabla = 1 \otimes d\right)$, $F^\bullet$

<u>Prop.</u> TFAE.

(1) $(\underline{V}, \mathcal{V}, \nabla, F^\bullet)$ satisfies Griffith transversality for every $V \in \mathrm{Rep}(G)$

(2) replace every by one almost faithful rep'n

(3) $V = \mathfrak{g}$ adjoint rep'n.

(4) Choose $\mu: \mathbb{G}_m \to G_{\mathbb{C}}$, $\mathfrak{g}_{\mathbb{C}} = \bigoplus_k \mathfrak{g}^k_{\mathbb{C}} = \mathfrak{g}^{-1}_{\mathbb{C}} \oplus \mathfrak{g}^0_{\mathbb{C}} \oplus \mathfrak{g}^1_{\mathbb{C}}$

(such μ is called minuscule cocharacter)

$$\begin{tikzcd} F^i\mathcal{V} \arrow[r, "\nabla"] \arrow[rr, bend right, "\mathcal{O}_S\text{-linear}"'] & \Omega_S \otimes \mathcal{V} \arrow[r] & \Omega_S \otimes \mathcal{V}/F^i\mathcal{V} \end{tikzcd}$$

$$T_S \otimes F^i\mathcal{V} \longrightarrow \mathcal{V}/F^i\mathcal{V}$$

$\mathrm{Lie}\, P_\mu = F^0_\mu \mathfrak{g}$

$$\mathfrak{g} \otimes \mathcal{O}_S / F^0(\mathfrak{g} \otimes \mathcal{O}_S) \simeq T_S$$

$\rho: G \to GL(V)$

$d\rho: \mathfrak{g} \otimes V \to V$

$$(\mathfrak{g} \otimes \mathcal{O}_S) \otimes \mathcal{V} \longrightarrow \mathcal{V}$$

$$F^i(\mathfrak{g} \otimes \mathcal{O}_S) \otimes F^j\mathcal{V} \longrightarrow F^{i+j}\mathcal{V}$$

$$\to \mathfrak{g} \otimes \mathcal{O}_S / F^0(\mathfrak{g} \otimes \mathcal{O}_S) \otimes F^i\mathcal{V} \longrightarrow \mathcal{V}/F^i\mathcal{V}$$

注。

$$F^i(\mathfrak{g}\otimes\mathcal{O}_S)/F^0(\mathfrak{g}\otimes\mathcal{O}_S)\otimes F^jV \longrightarrow F^{j-i}V/F^jV$$

If $\mathfrak{g}\otimes\mathcal{O}_S = F^{-1}(\mathfrak{g}\otimes\mathcal{O}_S)$, $\quad T_S\otimes F^iV \longrightarrow F^{i-1}V/F^iV$

$\Updownarrow$

μ minuscule. $\quad\Rightarrow$ Griffith transversality

$(\Leftarrow)$

<u>Lecture 10</u> $\quad G$ real (conn'd) reductive, μ conj. class of cocharacters of $G_{\mathbb{C}}$

$$\overset{G(\mathbb{R})}{\circlearrowright} D_{G,\mu} \subset \tilde{D}_{G,\mu} \subset \overset{G(\mathbb{C})}{\circlearrowright} \mathrm{Fil}_{G,\mu}$$

Assume $G = G_{ad}$, i.e. $Z_G = \{1\}$.

$x \in D_{G,\mu} \longleftrightarrow h_x : \mathbb{S} \longrightarrow G$

- $w_{h_x}(\mathbb{G}_m) \subset Z_G = \{1\}$ $\qquad h : \mathbb{S} \longrightarrow \mathbb{S}/w(\mathbb{G}_m) \simeq U(1) \longrightarrow G$
- $\mu_{h_x} \sim \mu$ $\qquad z \longmapsto z/\bar{z}$
- $\mathrm{Ad}\, h(i)$ is Cartan involution of G

$h(i)^2 = 1$

$G_h = \{g \in G(\mathbb{C}) : h(i)\bar{g}h(i)^{-1} = g\}$ is compact.

$K = G_h \cap G$ (as subgps of $G(\mathbb{C})$)

$= \{g \in G(\mathbb{R}) : g h(i) g^{-1} = h(i)\}$, K is max. cpt subgp of $G(\mathbb{R})$

$X = G(\mathbb{R})/K$ is called the symmetric space

$G(\mathbb{R})/K \simeq$ Set of Cartan involutions of $G(\mathbb{R})$

$X \overset{\text{analytically}}{\simeq} \mathbb{R}^n$ for some n.

$\left(\Rightarrow \pi_0(K) \simeq \pi_0(G(\mathbb{R})) \right)$

$\theta := \mathrm{ad}\, h(i) : \mathfrak{g} \to \mathfrak{g}$,

$$\mathfrak{g} = \overset{+1}{\mathfrak{k}} \oplus \overset{-1}{\mathfrak{p}}$$

- $\mathfrak{k}$ subalg, $\mathrm{Kil}|_{\mathfrak{k}}$ is negative def.
- $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$, $\mathrm{Kil}|_{\mathfrak{p}}$ is positive def
- $[\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p}$
- $G(\mathbb{R}) \simeq K \times P$

$k \exp(\eta) \longleftarrow\!\shortmid (k, \eta)$

$h = h_x$. $\quad Z_G(h)$ centralizer of $h(\mathbb{S})$ in G, quasi-Levi subgp of G.

$Z_G(h)(\mathbb{R}) \subset K$

$$D_{G,\mu} \longrightarrow X$$

simple G adjoint, μ minuscule $\longrightarrow$ $\mathbb{S} \hookrightarrow h \longmapsto h(i)$

$$G(\mathbb{R}) / Z_G(h)(\mathbb{R}) \to G/K$$

G adj., μ minuscule, $h \neq h^{-1}$

<u>Lemma</u> $\downarrow$ $D_{G,\mu} \cup D_{G,-\mu} \longrightarrow X$ is a double cover.

<u>Pf</u> $h(i)^2 = 1$, $h(i) = h(i)^{-1}$.

Let $h_1, h_2 \in D_{G,\mu} \cup D_{G,-\mu}$, $h_1(i) = h_2(i)$.

$h_i : \mathbb{S} \longrightarrow \mathbb{S}/w(G_m) \longrightarrow G$ $\qquad h_1(\mathbb{S}), h_2(\mathbb{S}) \subset K$

$\mathbb{S}/w(G_m) \cong U(1)$

~~Up to conj. by elts in~~ K,

If $j = 1, 2$, $h_j : \mathbb{S} \to T$ max. torus of K

$h_1 = h_2 \cdot h$ for some $h : \mathbb{S} \longrightarrow K$

$$\begin{tikzcd} \mathbb{S} \arrow[r] \arrow[dr] & K \\ & U(1) \arrow[u, "\bar{h}"'] \end{tikzcd}$$

$h(i) = 1$, $\bar{h}(-1) = 1$

$\mu_{h_1} = \mu_{h_2} + \mu_h$ ← divisible by 2

$\langle \mu_{h_j}, \alpha \rangle \in \{-1, 0, 1\}$ for any simple root of $G_{\mathbb{C}}$

$\mu_{h_1} \neq \mu_{h_2} \Rightarrow \mu_{h_1} = -\mu_{h_2}$.

<u>Pf</u> why $[h_1(\mathbb{S}), h_2(\mathbb{S})] = 1$:

When μ minuscule, $\mathrm{Lie}\, Z_G(h) = \mathrm{Lie}\, K$

$h(i)$ $\quad k \oplus p \quad \Rightarrow Z_G(h)^\circ = K^\circ$.

$$\mathfrak{g}_{\mathbb{C}} = \underbrace{\mathfrak{g}^{1,1} \oplus \mathfrak{g}^{0,0} \oplus \mathfrak{g}^{1,-1}}_{\mathfrak{p}_{\mathbb{C}} \quad \mathfrak{k}_{\mathbb{C}}}$$

$h_1(i) = h_2(i)$

$\Rightarrow h_1(\mathbb{S}) \subset K^\circ = Z_G(h_2)^\circ$

$\Rightarrow h_1(\mathbb{S}), h_2(\mathbb{S})$ are in the same max. torus T

<u>Rmk</u>.

μ in general, G in general

$\mathrm{Lie}\, Z_G(h)_{\mathbb{C}} = \mathfrak{g}^{0,0}$

$$T_h\left(G(\mathbb{R}) / Z_G(h)(\mathbb{R})\right) \xrightarrow{\sim} T_{\mu_h}\left(G_{\mathbb{C}} / P_\mu\right)$$

$\Rightarrow$ every $G(\mathbb{R})$-orbit is open in $D_{G,\mu}$ $\Rightarrow D_{G,\mu}$ is a union of $G(\mathbb{R})$-orbits.

Cor. (1) $G = G_{ad}$ simple, μ minuscule. then $D_{G,\mu}$ is conn'd if $\mu, -\mu$ are not conj.

$D_{G,\mu}$ has two conn'd components if $\mu, -\mu$ are conj.

(2) every conn'd component of $D_{G,\mu} \simeq X$.

Cor. $G = G_{ad}$ (simple), μ minuscule. then $G(\mathbb{R})$ acts transitively on $D_{G,\mu}$.

Pf can assume $\mu, -\mu$ are conjugate.

$G_h = \{ g \in G(\mathbb{C}) : h(i)\bar{g}h(i)^{-1} = g \}$ is compact

$h_1, h_2 \in D_{G,\mu} = D_{G,-\mu}$

$h_1(i) = h_2(i)$, $h_1 = h_2^{-1}$

$$\begin{tikzcd} h_1 : \mathbb{S} \arrow[r] & \mathbb{S}/w(G_m) \arrow[r] \arrow[d, phantom, "\wr"] & G \cap G_h = K \arrow[d] \\ & U(1) \arrow[r] & G_h \end{tikzcd}$$

μ_{h_1} conj. to μ_{h_2} in $G(\mathbb{C})$ $\Rightarrow$ h_1, h_2 conj. in G_h.

i.e. $\exists\, g \in G_h$, $\quad g h_1 g^{-1} = h_2$

$$g h_1(i) g^{-1} = h_2(i) = h_1(i)$$

$\Rightarrow g \in K \subset G(\mathbb{R})$ $\Rightarrow$ h_1, h_2 are conj. in $G(\mathbb{R})$.

Let $g \mapsto g^c = h(i)\bar{g}h(i)^{-1}$ be the cplx conj. of $G(\mathbb{C})$ w.r.t. G_h.

$G(\mathbb{R}) = \{ g \in G(\mathbb{C}) : h(i) g^c h(i)^{-1} = g \}$

Rmk. (Assume $G = G_{ad}$)

$$T_x D_{G,\mu} = (\mathfrak{g} \otimes \mathbb{C})/F^0 = \bigoplus_{p<0} \mathfrak{g}_{\mathbb{C}}^{-p,p}$$

μ is minuscule. $\simeq \mathfrak{g}_{\mathbb{C}}^{-1,1}$

(in fact, Kähler - Einstein metric)

$H(v,w) \geq \mathrm{Kil}_{\mathbb{C}}(v, \bar{w})$, $v, w \in \mathfrak{g}^{-1,1}$ is a hermitian metric on $D_{G,\mu}$.

$h(i)^2 = 1$. acts as -1 on $T_x D_{G,\mu}$. $\Rightarrow$ $D_{G,\mu}$ is a union of hermitian symmetric domains.

Rmk When G is not adjoint, $D_{G,\mu}$ may not be a homogeneous space of $G(\mathbb{R})$.

Ex. $G = U(1,1)$

$$D_{G,\mu} \ni \begin{cases} h_1 : z \longmapsto \begin{pmatrix} \frac{z}{\bar{z}} & \\ & 1 \end{pmatrix} \\ h_2 : z \longmapsto \begin{pmatrix} 1 & \\ & \frac{z}{\bar{z}} \end{pmatrix} \end{cases}$$

they are not conj. in $G(\mathbb{R})$

G simple. $G \to G_{ad}$

$$\begin{array}{ccc} D_{G,\mu} & \longrightarrow & D_{G_{ad},\mu_{ad}} \\ \downarrow & & \wr \\ \mathrm{Fil}_{G,\mu} & = & \mathrm{Fil}_{G_{ad},\mu_{ad}} \end{array}$$

in $PU(1,1)$, $\mu_{ad} \sim -\mu_{ad}$

Lecture 11 G real, μ. $D_{G,\mu} \underset{\text{open}}{\subset} \mathrm{Fil}_{G,\mu}(\mathbb{C})$

$$\begin{array}{ccc} D_{G,\mu} & & \\ \rotatebox{90}{$\in$} & & \\ h & & \\ G(\mathbb{R}) \cdot h & \xhookrightarrow{\text{open}} & D_{G,\mu} \end{array}$$

$HS_{\mathbb{R}}$, $\otimes$ - abelian cat $/\mathbb{R}$

$Kot_{\mathbb{R}}$: obj : - graded finite dim. $\mathbb{C}$-v.s. $V = \bigoplus V_m$

- $\varphi : V \longrightarrow V$ graded semilinear map s.t. $\varphi^2 = (-1)^m$ on V_m

$\otimes$ - abelian category

<u>Rmk</u>. $Kot_{\mathbb{R}}$ is an example of a non-neutral tannakian cat. $/\mathbb{R}$.

$\exists$ $\otimes$ - exact functor $HS_{\mathbb{R}} \longrightarrow Kot_{\mathbb{R}}$

$$(V, h) \longmapsto (V_{\mathbb{C}}, h(i)\,\overline{(\ )})$$

$$(h(i)^2 = h(-1) = (-1)^m)$$

<u>General construction</u>. k field

Let $\mathcal{T}$ be a k-linear tannakian cat. G conn'd reductive group $/k$.

<u>Def</u>. $Tor_G(\mathcal{T})$: G-torsors in $\mathcal{T}$

$$\overset{\shortparallel}{\{F : Rep_k(G) \to \mathcal{T} : \otimes\text{-exact (faithful) functors}\}}$$

$Tor_G(\mathcal{T})$ is a groupoid, isom. classes denoted by $H^1(\mathcal{T}, G)$

<u>Ex</u>. $\mathcal{T} = Vect_k$, $Tor_G(Vect_k)$ cat. of G-torsors $/k$

$\mathcal{T} = Vect_k$, $H^1(k, G)$, $BG(k)$

$$\begin{array}{ccc} F & & \text{is} \\ \downarrow & & \\ F(\mathcal{O}(G)) & \Big\{ & \text{schemes } X/k \text{ equipped w/ action } G \text{ s.t. } G \times_k X \xrightarrow[\sim]{(act,\, pr)} X \times_k X \Big\} \\ \uparrow & & \\ \text{left regular rep'n} & & \end{array}$$

$$\begin{array}{ccc} Vect_k & \hookrightarrow & \mathcal{T} \\ V & \longmapsto & V \otimes \mathbb{1}_{\mathcal{T}} \end{array} \rightsquigarrow Tor_G(Vect_k) \longrightarrow Tor_G(\mathcal{T})$$

$b : \mathrm{Rep}_k(G) \longrightarrow \mathcal{T}$

$J_b = \mathrm{Aut}(b)$ $\quad$ k-group

$J_b(R) = \mathrm{Aut}^{\otimes}(b_R)$, $b_R : \mathrm{Rep}(G) \otimes_k R \longrightarrow \mathcal{T} \otimes_k R$.

If $\mathcal{T} = \mathrm{Vect}_k$, $b : \mathrm{Rep}(G) \longrightarrow \mathrm{Vect}_k$ is usually written as ξ, $J_b = G^{\xi}$

called a pure inner form of G.

In general, choose $\omega : \mathcal{T} \longrightarrow \mathrm{Vect}_{k'}$

$$\mathrm{Rep}_k(G) \xrightarrow{b} \mathcal{T} \xrightarrow{\omega} \mathrm{Vect}_{k'}$$

$$(J_b)_{k'} = \mathrm{Aut}^{\otimes}(b)_{k'} \hookrightarrow \mathrm{Aut}^{\otimes}(\underset{\xi}{\underset{\shortparallel}{\omega \circ b}}) = (G_{k'})^{\xi}$$

$$(J_b)_{\bar{k}} \hookrightarrow G_{\bar{k}}$$

If this is an isom. we call b basic $\rightsquigarrow$ $\mathrm{Tor}_G(\mathcal{T})_{bsc} \subset \mathrm{Tor}_G(\mathcal{T})$

$H^1(\mathcal{T}, G)_{bsc} \subset H^1(\mathcal{T}, G)$

$\mathrm{Tor}_G(\mathrm{Vect}_k) \subset \mathrm{Tor}_G(\mathcal{T})_{bsc} \subset \mathrm{Tor}_G(\mathcal{T})$

<u>Ex.</u> $\mathcal{T} = \mathrm{Rep}_k(\mathbb{D})$, $\mathbb{D}$ (pro-) alg group $/k$. Then objects in $\mathrm{Tor}_G(\mathcal{T})$ can be represented by $b = (\xi_b, \nu_b)$, $\nu_b : \mathbb{D} \longrightarrow G^{\xi_b}$ hom.

$$H^1(\mathcal{T}, G) = \coprod_{\xi \in H^1(k,G)} \mathrm{Hom}(\mathbb{D}, G^{\xi}) /\!/ G^{\xi}(k). \qquad J_b = Z_{G^{\xi}}(\nu_b)$$

$$\mathrm{Tor}_G(T)_{bsc} = \{(\xi_b, \nu_b) : \nu_b : \mathbb{D} \to Z_G\} \supset \mathrm{Tor}_G(\mathrm{Vert}_k)$$

$$\mathrm{Tor}_G(\mathrm{Vert}_k) = \{(\xi_b, \nu_b) : \nu_b \text{ is trivial}\}$$

$$\begin{tikzcd}
H^1(T, G) \arrow[r] \arrow[dr, "\nu_G"'] & H^1(T_{\bar{k}}, G_{\bar{k}}) \arrow[d] \\
& \mathrm{Hom}(\mathbb{D}, G_{\bar{k}}) /\!/ G_{\bar{k}} \arrow[u]
\end{tikzcd}$$

This is called the Newton map.

$[\nu_b = \nu_G(b) : \mathbb{D} \to G_{\bar{k}}$ up to conj. is defined over $k]$

b basic $\iff$ $\nu_b = \nu_G(b)$ factors through the centre.

$b \in H^1(k, G)$ iff ν_b is trivial

If $G = G_{ad}$, $H^1(k, G) = H^1(T, G)_{bsc}$.

In general, $G \to G_{ad}$

$$\begin{tikzcd}
H^1(k, G) \arrow[r, hook] \arrow[rr, bend right] & H^1(T, G)_{bsc} \arrow[r] & H^1(T, G_{ad})_{bsc} \\
& & H^1(k, G_{ad}) \arrow[u, "\wr"']
\end{tikzcd}$$

Let $b_1, b_2 \in \mathrm{Tor}_G(T)$, $J_{b_1, b_2} := \mathrm{Isom}^{\otimes}(b_1, b_2)$ k-scheme.

pseudo $J_{b_1} \times J_{b_2}$-bi torsor

And this is a J_{b_2}-torsor iff $\nu_{b_1} = \nu_{b_2}$.

$$H^1(k, J_b) \longrightarrow H^1(\tau, G)$$
$$\xi \longmapsto b \overset{J_b}{\times} \xi$$

is surj. on fibre of $H^1(\tau, G) \xrightarrow{\nu_G} \mathrm{Hom}(\mathbb{D}, G_{\bar{k}})/G_{\bar{k}}$ containing b.

$$(V, h) \longmapsto (V_{\mathbb{C}}, h(i)(-))$$

$$\tau_1 = HS_{\mathbb{R}} \longrightarrow \tau_2 = \mathrm{Kot}_{\mathbb{R}}$$

$$H^1(HS_{\mathbb{R}}, G) = \coprod_{\xi \in H^1(\mathbb{R}, G)} \mathrm{Hom}(\mathbb{S}, G^{\xi})/G^{\xi}(\mathbb{R}) \qquad (\xi, h)$$

$$\downarrow \qquad\qquad \downarrow$$

$$H^1(\mathrm{Kot}_{\mathbb{R}}, G) =: B(G) \qquad V \longmapsto V \otimes_{\mathbb{R}} \mathbb{C} \qquad (\omega_h, h(i))$$

$$- \quad (\mathrm{Kot}_{\mathbb{R}})_{\mathbb{C}} \simeq \mathrm{Vect}_{\mathbb{C}}^{gr}$$

$$\mathbb{D} = \mathbb{G}_m$$

$$- \quad B(G) = \{(\nu : \mathbb{G}_m \to G_{\mathbb{C}},\ g \in G(\mathbb{C})) : g\bar{g} = \nu(-1)\} / G(\mathbb{C})$$

$$(\nu, g) \sim (h\nu h^{-1}, hg\bar{h}^{-1})$$

$$H^1(\mathbb{R}, G) \longrightarrow X^*(\pi_0(Z_{\hat{G}}^{\Gamma_{\mathbb{R}}})) \quad h \in G(\mathbb{C})$$

When $\tau = \mathrm{Kot}_{\mathbb{R}}$, $\exists$ a map (Kottwitz map) $k_G : H^1(\mathrm{Kot}_{\mathbb{R}}, G) \longrightarrow X^*(Z_{\hat{G}}^{\Gamma_{\mathbb{R}}})$

$$\| $$

$$\pi_1(G_{\mathbb{C}})_{\Gamma_{\mathbb{R}}} = (X_*(T_{\mathbb{C}})/\text{coroots})_{\Gamma_{\mathbb{R}}}$$

When $G = T_{\mathbb{R}}$, k_G is an isom.

In general, $T \longrightarrow G$

$$\begin{array}{ccc} H^1(\mathrm{Kot}_{\mathbb{R}}, T) & \xrightarrow{\kappa_T} & X^*(\hat{T}^{\Gamma_{\mathbb{R}}}) \\ \downarrow & & \downarrow \\ H^1(\mathrm{Kot}_{\mathbb{R}}, G) & \xrightarrow{\kappa_G} & X^*(Z_{\hat{G}}^{\Gamma_{\mathbb{R}}}) \end{array}$$

<u>Lemma</u>. Let G be a conn'd reductive group $/\mathbb{R}$,

$B(Z_G) \curvearrowright B(G)_{bsc}$ preserving fibers of $B(G)_{bsc} \longrightarrow B(G_{ad})_{bsc}$.

$$B(G_{ad})_{bsc} = H^1(\mathbb{R}, G_{ad})$$

Suppose $\exists$ cpt inner form G_c of G_{ad}.

Let $B(G)_{cpt}$ be its preimage in $B(G)_{bsc}$.

Then $B(Z_G)$ acts simply-transitively along $B(G)_{cpt}$.

<u>Rmk</u>. $B(Z_G)$ acts transitively on other fibres, but not simply transitively in general.

$$H^1(HS_{\mathbb{R}}, G) = \coprod_{\xi \in H^1(\mathbb{R}, G)} \mathrm{Hom}(\mathbb{S}, G^{\xi}) / G(\mathbb{R})$$

$$\cup$$

$$H^1(HS_{\mathbb{R}}, G)^{w.p.} = \left\{ h : \begin{array}{l} w_h(\mathbb{G}_m) \subset Z_G \\ \mathrm{Ad}\, h(i) \text{ is a Cartan inv'n of } G_{ad}^{\xi} \end{array} \right\}$$

$$\left(\coprod_{\mu} D_{G,\mu} / G(\mathbb{R}) \quad \xi = 1 \right)$$

$$\begin{array}{ccc} h & \longmapsto & (w_h, h(i)) \\ H^1(HS_{\mathbb{R}}, G)^{w.p.} & \longrightarrow & B(G)_{cpt} \\ h & \longmapsto & \mu_h \in \mathrm{Hom}(\mathbb{G}_m, G_{\mathbb{C}}) /\!/ G_{\mathbb{C}} \end{array}$$

<u>Claim</u>: $H^1(HS_{\mathbb{R}}, G)^{w.p.} \to \mathrm{Hom}(\mathbb{G}_m, G_{\mathbb{C}}) /\!/ G_{\mathbb{C}} \times B(G)_{cpt}$ is injective.

Image are those (μ, b):

$$k_G(b) - [\mu] \in X^*(\pi_0(Z_G^{\Gamma_{\mathbb{R}}})) \subset X^*(Z_G^{\Gamma_{\mathbb{R}}})$$

Lecture 12

$$\begin{array}{c} X \\ f\downarrow \\ \text{connected} \to S \end{array}$$

smooth proj. morphism $/\mathbb{C}$ of smooth var.

$$\left(R^i f_* \Lambda_{X^{an}}/\text{tor},\ \mathcal{H}^i_{dR}(X^{an}/S^{an}),\ \nabla^{GM},\ F^i\right)$$

$\mathbb{L}$, rank $\mathbb{L} =: d$

$$\begin{array}{c} S \\ \cup \\ s \end{array} \longrightarrow D_{\mathbb{L}_s,(d_i)}$$

doesn't extend to $S \to D_{V,(d_i)}$ because $R^i f_* \Lambda_{X^{an}}/\text{tor}$ is not trivial.

$$\begin{tikzcd}
 & (s, \mathbb{L}_s \simeq \Lambda^d) \arrow[r, bend left] & \mathrm{Fil}_{V,(d_i)} \\
\Gamma \arrow[d, equal] & \tilde{S} \arrow[r] \arrow[d] & D_{V,(d_i)} \arrow[d] \\
GL_d(\Lambda) \arrow[r] & S^{an} \arrow[r] & \Gamma \backslash D_{V,(d_i)}
\end{tikzcd}$$

$\tilde{S} \subset \mathrm{Fil}_{V,(d_i)}$

$GL_d(\Lambda)$ - torsor

Def'n Let G be a group. $\Gamma_1, \Gamma_2 \subset G$ are called commensurable if

$$\# \Gamma_1/\Gamma_1 \cap \Gamma_2,\ \# \Gamma_2/\Gamma_1 \cap \Gamma_2 < \infty.$$

Lemma (1) Commensurability is an equivalence relation on the set of subgps of G

(2) $f: H \to H'$, $\Gamma_1, \Gamma_2 \subset H$ commensurable, then $f(\Gamma_1), f(\Gamma_2)$ commensurable

(3) $f: H \to H'$, $\Gamma_1, \Gamma_2 \subset H'$ com., then $f^{-1}(\Gamma_1), f^{-1}(\Gamma_2)$ com.

Now let G be a linear algebraic group $/\mathbb{Q}$, $\mathcal{G}_1, \mathcal{G}_2$ two affine flat models of $G/\mathbb{Z}$

(e.g. $G \hookrightarrow GL_n$ & takes $\mathcal{G}$ to be the Zariski closure)

<u>Lemma</u>. $\mathcal{G}_1(\mathbb{Z}), \mathcal{G}_2(\mathbb{Z}) \subset G(\mathbb{Q})$ are commensurable.

<u>Pf</u>. $\exists N$, $\mathcal{G}_1|_{\mathbb{Z}[\frac{1}{N}]} \simeq \mathcal{G}_2|_{\mathbb{Z}[\frac{1}{N}]}$

$$\begin{tikzcd}
\mathcal{G}_i(\mathbb{Z}) \arrow[r] \arrow[d] & \mathcal{G}_i(\mathbb{Z}[\tfrac{1}{N}]) \arrow[r] \arrow[d] & G(\mathbb{Q}) \\
\prod_{p|N} \mathcal{G}_i(\mathbb{Z}_p) \arrow[r] & \prod_{p|N} G(\mathbb{Q}_p) &
\end{tikzcd}$$

But $\mathcal{G}_1(\mathbb{Z}_p), \mathcal{G}_2(\mathbb{Z}_p)$ are open cpt subgps of $G(\mathbb{Q}_p)$ ← locally compact

So $\mathcal{G}_1(\mathbb{Z}_p), \mathcal{G}_2(\mathbb{Z}_p)$ are commensurable in $G(\mathbb{Q}_p)$.

<u>Def</u>. Let G be a reductive alg. gp $/\mathbb{Q}$. A <u>congruence subgp</u> of G is a subgp of $G(\mathbb{Q})$ of the form $\mathcal{G}(\mathbb{Z})$ for some affine flat model of G over $\mathbb{Z}$.

<u>Def</u> $k \supset \mathbb{Q}$ (e.g. $k = \mathbb{Q}, \mathbb{R}$) A subgp Γ of $G(k)$ is called an <u>arithmetic subgp</u> if Γ is commensurable w/ some congruence subgp of G.

<u>Rmk</u>. $K_p = \mathcal{G}(\mathbb{Z}_p) \subset G(\mathbb{Q}_p)$

$K = \prod K_p \subset G(\mathbb{A}_f)$, $\mathbb{A}_f = \prod_{p}' \mathbb{Q}_p$ finite adele.

then $\mathcal{G}(\mathbb{Z}_p) = G(\mathbb{Q}) \cap K$ inside $G(\mathbb{A}_f)$

Conversely, given any $K \underset{\text{open cpt}}{\subset} G(\mathbb{A}_f)$, $\exists\, \mathcal{G}/\mathbb{Z}$ s.t. $K_p = \mathcal{G}(\mathbb{Z}_p)$. ($\mathcal{G}$ may not be unique)

So congruence subgps can also be defined as $G(\mathbb{Q}) \cap K$.

<u>Ex.</u> (1) $G = \mathbb{G}_m$, congruence subgp $\{1\}$ or $\{\pm 1\}$.

In general, $G = \mathrm{Res}_{E|\mathbb{Q}}\, \mathbb{G}_m$, E no. field,

congruence subgps = arithmetic subgps are finite index subgps of $\mathcal{O}_E^\times$.

(2) $G = SL_2$,

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) : \begin{matrix} a \equiv d \equiv 1 \\ b \equiv c \equiv 0 \end{matrix} \mod N \right\}$$

$$\cap$$

$$\Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) : \begin{matrix} a \equiv d \equiv 1 \\ c \equiv 0 \end{matrix} \mod N \right\}$$

$$\cap$$

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) : c \equiv 0 \mod N \right\}$$

congruence subgps

Finite index subgps of $SL_2(\mathbb{Z})$ are arithmetic, but not necessarily congruence.

<u>Def'n</u> An elt $\gamma \in G(k)$ is <u>neat</u> if for every alg. rep'n (V, ρ) of G/k, the eigenvalues of $\rho(\gamma)$ generate a torsion free subgp of $\bar{k}^\times$.

A subgp $\Gamma \subset G(k)$ is <u>neat</u> if every elt in Γ is neat.

<u>Lemma</u> Every arithmetic subgp of $G(k)$ contains a neat subgroup of finite index (hence also arithmetic)

<u>Pf.</u> We can assume $\Gamma = G(k) \cap GL_n(\mathbb{Z})$ for some $G \hookrightarrow GL_n$.

Let M be the set of cyclotomic polynomials of degree $\leq n!$. $p \nmid f(1)$, $\forall f \in M$.

$\Gamma(p) = \{ g \in GL_n(\mathbb{Z}) : g \equiv 1 \mod p \}$; $\Gamma \cap \Gamma(p)$ finite index in Γ

$\gamma \in \Gamma(p)$, $\quad \mathrm{char}(\gamma, t)$

$\subset E$ splitting field $\quad [E:\mathbb{Q}] \le n!$

If $E \ni \zeta$ (root of $\mathrm{char}(\gamma,t)$) is a root of unity, $\gamma \equiv 1 \mod p$. $\Rightarrow \zeta \equiv 1 \mod p$

$\Rightarrow p \mid \mathrm{char}(\zeta, 1)$, contradiction.

<u>Rmk</u> $\Gamma(N) \subset GL_n(\mathbb{Z})$ is torsion-free for $N \geq 3$.

$g^m = 1$

$\zeta_1, \cdots, \zeta_n$, $\quad \zeta_i \equiv 1 \mod N$ $\quad . N \mid \zeta - 1$

$N \geq 3$, $\quad \zeta$ root of 1 $\quad$, contradiction

Lemma (1) Neat subgps are torsion free

(2) Let $K \subset G(\mathbb{R})$ be a compact subgp, & $\Gamma \subset G(\mathbb{R})$ neat, then $\Gamma \cap K = \{1\}$

(3) If $f: G \to G'$ is a hom. Then f sends neat subgps of $G(k)$ to neat subgps of $G'(k)$.

<u>Lemma</u> arithmetic subgps Γ of $G(\mathbb{R})$ are discrete.

<u>Pf</u> $\Gamma = G(\mathbb{R}) \cap GL_n(\mathbb{Z})$, $G \hookrightarrow GL_n$

<u>Prop</u> Let $f: G \to G'$ be a hom. of $\mathbb{Q}$-gp

(1) If f is <u>injective</u>, $\Gamma \subset G'(k)$ arithmetic, then $f^{-1}(\Gamma)$ is arithmetic.

(2) If f is <u>surjective</u>, $\Gamma \subset G(k)$ arithmetic, then $f(\Gamma) \subset G'(k)$ arithmetic.

Pf (1) $G \hookrightarrow G' \hookrightarrow GL_n$

$$G(k) \cap GL_n(\mathbb{Z}) \hookrightarrow G'(k) \cap GL_n(\mathbb{Z})$$

(2)
$$\begin{tikzcd} K \arrow[r, hook] & G \arrow[r, two heads, "f"] \arrow[d, hook] & G' \\ & GL_n & \end{tikzcd}$$

$$\begin{tikzcd} \mathcal{K} \arrow[r, phantom, "\subset"] & \mathcal{G} \arrow[r, "\tilde{f}"] \arrow[d, hook] & \mathcal{G}' \\ & GL_n & \end{tikzcd}$$

$$\mathcal{G}(\mathbb{Z}) \longrightarrow \mathcal{G}'(\mathbb{Z}) \longrightarrow H^1_{fppf}(\mathbb{Z}, \mathcal{K})$$

$$\mathcal{G}'(\mathbb{Z}) / f(\mathcal{G}(\mathbb{Z})) \hookrightarrow H^1_{fppf}(\mathbb{Z}, \mathcal{K})$$

Raynaud:

G/H = scheme ? (a priori, exists as algebraic space)

over general regular noeth. base, still open.

over Dedekind domain ✓

2-dim'l base ✓ [Pappas-Zhu]

BIG

Thm. Let $\mathcal{G}$ be an affine flat group scheme of finite type $/\mathcal{O}_{E,S}$ ← ring of S-integers of some # field E

then $H^1_{fppf}(\mathcal{O}_{E,S}, \mathcal{G})$ is finite!

Ex. (1) $\mathcal{G} = \mathbb{G}_m / \mathcal{O}_E$. $H^1(\mathcal{O}_E, \mathbb{G}_m) = Cl(\mathcal{O}_E)$

Essential case: $\mathcal{G}$ affine smooth connected fibre $/\mathcal{O}_E$ $(= \mathbb{Z})$ [Borel, Harish-Chandra]

Lemma. $\mathcal{G}$ as above, $\exists$ equiv. of groupoids

$$\{\mathcal{G}\text{-torsors over } \mathbb{Z}\} \xrightarrow{\sim} \coprod_{\xi \in Ker^1_f(\mathbb{Q}, G)} G^\xi(\mathbb{Q}) \backslash G(\mathbb{A}_f) / \mathcal{G}(\hat{\mathbb{Z}}), \quad G = \mathcal{G}_{\mathbb{Q}}$$

where $Ker^1_f(\mathbb{Q}, G) = \ker\left(H^1(\mathbb{Q}, G) \longrightarrow \prod H^1(\mathbb{Q}_p, G)\right)$

Tate-Shafarevich group $Ш_f(\mathbb{Q}, G)$

Thm $\Leftrightarrow$ $\# \, G^{\mathfrak{z}}(\mathbb{Q}) \backslash G(\mathbb{A}_f) / \mathfrak{g}(\hat{\mathbb{Z}}) < \infty$

& $\ker^1_{\mathfrak{g}}(\mathbb{Q}, G) < \infty$.

$$\begin{array}{ccccccc} \mathcal{P}_{\mathbb{F}_p} & & \mathcal{P}_{\mathbb{Z}_p} & & \mathcal{P} & & \mathcal{P} \\ \downarrow & & \downarrow & & \downarrow & & \\ \operatorname{Spec} \mathbb{F}_p & \to & \operatorname{Spec} \mathbb{Z}_p & \to & \operatorname{Spec} \mathbb{Z} & \leftarrow & \operatorname{Spec} \mathbb{Q} \end{array}$$

<u>Lecture 13</u>. Let $G = G_{ad} / \mathbb{R}$, $\mathbb{R}^d \approx D \subset D_{G,\mu} \simeq G_{ad}/K$ be a conn'd component of a <u>classical</u> MT domain. Let $\Gamma \subset G(\mathbb{R})^+$ be an torsion-free arithmetic subgroup. Then the action of Γ on D is free, <u>proper</u>.

Ψ
x

$x \in D$, $\exists$ U open nbhd of x s.t.

$\Gamma \cap G_x = \{1\}$. $\quad \gamma(U) \cap U = \phi$, $\forall \gamma \in \Gamma$.

$G(\mathbb{R})^+$ neutral conn'd component of $G(\mathbb{R})$ w.r.t. real topology.

$D \xrightarrow{\pi} \Gamma \backslash D$ local homeomorphism
covering map

So $\Gamma \backslash D$ has a unique cplx mfld str. s.t. π is holomorphic.

<u>Thm.</u> (Baily-Borel, Borel) Let (G, D, Γ) be as above. Then ① $\exists$ <u>smooth</u> quasi-proj. variety $D(\Gamma)$ (quasi-proj.) over $\mathbb{C}$ w/ $D(\Gamma)^{an} = \Gamma \backslash D$.

② for every smooth $\mathbb{C}$-var. S,

$$\operatorname{Hom}_{\mathrm{Var}_{\mathbb{C}}}(S, D(\Gamma)) \xrightarrow{\sim} \operatorname{Hom}_{\text{cplx mfld}}(S^{an}, D(\Gamma)^{an})$$

(3) $D(\Gamma) \xhookrightarrow{\text{open}} D(\Gamma)^*$ w/ $D(\Gamma)^*$ normal projective. s.t. for every smooth var. X w/ a normal crossing divisor Z, and for every

$$\begin{tikzcd} X \setminus Z \arrow[r] \arrow[d, hook] & D(\Gamma) \arrow[d, hook] \\ X \arrow[r, dashed, "\exists!"] & D(\Gamma)^* \end{tikzcd}$$

<u>Rmk.</u> (2) implies that the alg. str. on $\Gamma \backslash D$ is unique

$D(\Gamma)$ is unique: $D(\Gamma)'$ another sm. alg. var. $(D(\Gamma)')^{an} \simeq \Gamma \backslash D$,

$D(\Gamma)' \xrightarrow{\sim} D(\Gamma)$.

(3) $\Rightarrow$ (2)

$$\begin{tikzcd} S \arrow[d, hook, "\text{resol'n of sing.}"'] \\ X \end{tikzcd}$$

projective ~~proper~~ smooth

$X - S$ normal crossing divisor

$$\begin{tikzcd} S^{an} \arrow[r] \arrow[d] & \Gamma \backslash D \arrow[d] \\ X^{an} \arrow[r, dashed] & (D(\Gamma)^*)^{an} \end{tikzcd}$$

use GAGA.

Essential ingredients: existence of $D(\Gamma)^*$, and extension property.

Existence of $D(\Gamma)^*$

Case $\Gamma \backslash D$ is compact: On D we have $\underbrace{\text{hermitian metric}}_{\text{positive curvature}}$ (Kähler)

$(\Gamma \backslash D, \omega_{\Gamma \backslash D})$ ← positive hermitian metric

algebraic by Kodaira embedding:

$\Gamma \backslash D \hookrightarrow \mathbb{P}^n$, so $\Gamma \backslash D = D(\Gamma)^{an}$ for some proj. var.

In this case $D(\Gamma)^* = D(\Gamma)$.

Then one still needs to show the extension property.

$$\begin{tikzcd} D^\times \arrow[r] \arrow[d, leftrightarrow] & \Gamma\backslash D \\ \mathbb{D} \arrow[ur, dashed] & \end{tikzcd}$$

hyperbolicity

$\dim_{\Gamma\backslash D} > 1$.

In general, $A = \bigoplus_{n \geq 0} H^0(\Gamma\backslash D, \omega^{\otimes n})$ is fin. gen. $\mathbb{C}$-alg,

and $D(\Gamma)^* = \mathrm{Proj}(A)$.

$$(D(\Gamma)^*)^{an} = \Gamma\backslash D^* = D \cup \cdots$$

eg. $D = \mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}\, z > 0\}$

$\Gamma \subset \mathrm{Im}(SL_2(\mathbb{Z}) \to PGL_2(\mathbb{R}))$

$\Gamma\backslash D$ is <u>not compact</u>

$D^* = \mathbb{H} \cup \mathbb{P}^1(\mathbb{Q}) = \mathbb{H}^*$, $\Gamma\backslash D^*$ smooth proj. curve

$\Gamma\backslash D^* - \Gamma\backslash D$ are called cusps.

Deligne: $\exists\, \Gamma$,

$$\begin{tikzcd} \Gamma\backslash D \arrow[r, equal] \arrow[d] & \mathbb{P}^1 \setminus \{0, 1, \infty\} \arrow[d] \\ \Gamma\backslash D^* \arrow[r, equal] & \mathbb{P}^1 \end{tikzcd}$$

Picard's big theorem

Cor G in general, $G(\mathbb{R})$-orbit $D \subset D_{G,\mu}$ ($\leftrightarrow \mu$ minuscule) $\leftarrow$ classical

$\Gamma \subset G(\mathbb{R})$ arithmetic, <u>neat</u>, $\Gamma\backslash D = D(\Gamma)^{an}$ for some unique quasi-proj. var.

$$\begin{array}{ccc} \Gamma & \longrightarrow & \pi(\Gamma) \text{ arith, neat} \\ \cap & & \\ G(\mathbb{R}) & & \end{array}$$

$$\begin{tikzcd} G \arrow[r, "\pi"] & G_{ad} \end{tikzcd} \qquad \begin{tikzcd} D \arrow[r, hook] \arrow[d, leftrightarrow] & D_{G_{ad},\mu} \arrow[d, leftrightarrow] \\ \mathrm{Fil}_{G,\mu} \arrow[r, equal] & \mathrm{Fil}_{G_{ad},\mu_{ad}} \end{tikzcd}$$

A Shimura datum (G, X) consists of

<u>Def</u> — G a (cont'd) reductive $\mathbb{Q}$-group.

— X a $G(\mathbb{R})$-conj. class of hom. $h : \mathbb{S} \to G_{\mathbb{R}}$ s.t.

X is a $G(\mathbb{R})$ orbit of a classical MT domain $D_{G,\mu}$

(SV1) $\mathbb{S} \xrightarrow{h} G_{\mathbb{R}} \xrightarrow{Ad} GL(\mathfrak{g}_{\mathbb{R}})$ defines a pure $\mathbb{R}$-HS w/ Hodge decomposition

$$\mathfrak{g}_{\mathbb{C}} \simeq \mathfrak{g}_{\mathbb{C}}^{-1,1} \oplus \mathfrak{g}_{\mathbb{C}}^{0,0} \oplus \mathfrak{g}_{\mathbb{C}}^{1,-1}$$

(SV2) $Ad\, h(i)$ is a Cartan involution of $(G_{ad})_{\mathbb{R}}$.

$$X \longleftrightarrow \{(\mu, \overset{\overset{B(G_{\mathbb{R}})}{\cup}}{b})\} \text{ satisfying certain condition}$$

Now let K be an open cpt subgp of $G(\mathbb{A}_f)$

$$G(\mathbb{Q}) \backslash X \times G(\mathbb{A}_f) / K \longrightarrow G(\mathbb{Q}) \backslash G(\mathbb{A}_f) / K \leftarrow \text{finite set}$$

$$\bigsqcup_{g_i} G(\mathbb{Q}) \backslash \underset{\Gamma_i \backslash X}{\underset{\cong}{X \times G(\mathbb{Q}) g_i K / K}}$$

$\{g_i\}$ set of representatives

$\Gamma_i = G(\mathbb{Q}) \cap g_i K g_i^{-1}$.

$\uparrow$ congruence subgp of $G(\mathbb{Q})$

<u>Def</u> K is called <u>neat level</u> if each Γ_i is neat.

<u>Lemma</u> Given any K, $\exists\, K' \triangleleft K$ open cpt. s.t. $\Gamma_i = G(\mathbb{Q}) \cap g_i K' g_i^{-1}$ is neat for every double coset $G(\mathbb{Q}) g_i K'$.

<u>Pf</u> $G(\mathbb{A}_f) = \bigsqcup G(\mathbb{Q}) h_j K$. We choose $K' \triangleleft K$ s.t. $G(\mathbb{Q}) \cap h_j K' h_j^{-1}$ is neat. $K = \bigsqcup k_i K'$. $G(\mathbb{A}_f) = \bigcup G(\mathbb{Q}) h_j k_i K'$.

$G(\mathbb{Q}) \cap h_j k_i K' k_i^{-1} h_j^{-1} = G(\mathbb{Q}) \cap h_j K' h_j^{-1}$ is neat!

Prop/Def. Let (G,X) be a Shimura datum, $K \subset G(\mathbb{A}_f)$ be a <u>neat</u> level.

$\exists!$ quasi-proj. var. $Sh_K(G,X)_{\mathbb{C}}$ over $\mathbb{C}$ s.t.

$$Sh_K(G,X)^{an}_{\mathbb{C}} \simeq G(\mathbb{Q})\backslash X \times G(\mathbb{A}_f)/K$$

For general level K, $K \supset K'$ neat

$$Sh_{K'}(G,X)_{\mathbb{C}} \circlearrowleft K/K'$$

$$Sh_K(G,X)_{\mathbb{C}} := Sh_{K'}(G,X)_{\mathbb{C}}/(K/K')$$

<u>Lecture 14</u>. (G,X) Shimura datum (SV1) (SV2)

$K \subset G(\mathbb{A}_f)$ (neat) level

$Sh_K(G,X)_{\mathbb{C}}$ complex quasi-proj. var.

$$Sh_K(G,X)^{an}_{\mathbb{C}} = G(\mathbb{Q})\backslash X \times G(\mathbb{A}_f)/K$$

$$\simeq \coprod_i \Gamma_i\backslash X$$

$$G(\mathbb{A}_f) = \coprod_i G(\mathbb{Q})\, r_i K, \qquad \Gamma_i = G(\mathbb{Q}) \cap r_i K r_i^{-1},$$

$\bar{\Gamma}_i$ = Image of Γ_i in $\left(G(\mathbb{Q}) \to G_{ad}(\mathbb{Q})\right)$

<u>Lemma</u>. $\exists$ a canonical bijection between $Sh_K(G,X)(\mathbb{C})$ and the set of triples,

(E,h,η) :
- E <u>trivialisable</u> G-torsor $/\mathbb{Q}$
- $h: \mathbb{S} \to \mathrm{Aut}(E)_{\mathbb{R}} \simeq G_{\mathbb{R}}$ belonging to X
- $\eta: E^0|_{\mathbb{A}_f} \xrightarrow{\sim} E|_{\mathbb{A}_f}$ is a K-orbit of trivialisations.

If $K = \mathcal{G}(\hat{\mathbb{Z}})$, then ($\mathcal{G}$ smooth, fibrewise connd $/\mathbb{Z}$)

$$Sh_K(G,X)(\mathbb{C}) \simeq \left\{ (\mathcal{E}, h) : \begin{array}{l} (1)\ \mathcal{E}: \mathcal{G}\text{-torsor on } \operatorname{Spec}\mathbb{Z},\ (2)\ \mathcal{E}_{\mathbb{Q}} \text{ is trivialisable} \\ (3)\ (h: \mathbb{S} \to \operatorname{Aut}(\mathcal{E})_{\mathbb{R}}) \in X \end{array} \right\}$$

<u>Rmk</u>. Can define

$$\widetilde{Sh_K(G,X)_{\mathbb{C}}} = \{ (\mathcal{E}, h) : (1), (3) \}$$

$$= \coprod_{\xi \in \ker^1(\mathbb{Q}, G)} Sh_K(G^{\xi}, X)_{\mathbb{C}}$$

<u>Additional axioms</u>.

(SV2') $\operatorname{Ad} h(i)$ is a Cartan involution of $G_{\mathbb{R}}/w_h(\mathbb{G}_m)$

(SV3) Any $\mathbb{Q}$-simple factor of G_{ad} is of non-cpt type

(SV4) The weight hom. is def'd $/\mathbb{Q}$

(SV5) The group $Z^{\circ}_G(\mathbb{Q})$ is discrete in $Z^{\circ}_G(\mathbb{A}_f)$

(SV6) Z°_G splits over a CM field.

<u>Rmk</u>. (SV2') $\Rightarrow$ (SV5)

<u>Prop</u>. Let T be a torus $/\mathbb{Q}$. TFAE

(i) $T(\mathbb{Q})$ is discrete in $T(\mathbb{A}_f)$

(ii) Every arithmetic subgp of $T(\mathbb{R})$ is finite

(iii) Let $T^a = \bigcap_{\chi} \ker(\chi: T \to \mathbb{G}_m)$, then $T^a(\mathbb{R})$ is cpt

(iv) $X^*(T)^{\mathrm{Gal}(\bar{\mathbb{Q}}|\mathbb{Q})} = X^*(T)^{\mathrm{Gal}(\mathbb{C}|\mathbb{R})}$

Eg. $T = \mathrm{Res}_{F/\mathbb{Q}} \mathbb{G}_m$, F totally real, $[F:\mathbb{Q}] > 1$

Then equiv. conditions fail.

$\# \mathcal{O}_F^\times = \infty$.

$T^a = \{t : N_m(t) = 1\}$

$T(\mathbb{R}) = (\mathbb{R}^\times)^{[F:\mathbb{Q}]}$

$\not\cup$

$T^a(\mathbb{R})$

$\mathrm{Res}_{F/\mathbb{Q}} GL_2$ (SV5) fails

Ex. G is MT gp of (V,h), $X \ni h : \mathbb{S} \to G_{\mathbb{R}}$. Then (SV2'), (SV4) hold

$\Downarrow$

(SV5).

Rmk (SV5) $\Rightarrow$ Z_G° splits over a CM field (SV6)

Why (SV5) is important?

$\Gamma_i = G(\mathbb{Q}) \cap r_i K r_i^{-1}$

$\downarrow$

$X \curvearrowleft \overline{\Gamma_i} \subset G_{ad}(\mathbb{Q})$

$\ker(\Gamma_i \to \overline{\Gamma_i}) = Z_G(\mathbb{Q}) \cap r_i K r_i^{-1}$.

When (SV5) holds, can choose K small enough (neat), $Z_G(\mathbb{Q}) \cap r_i K r_i^{-1} = \{1\}$

Γ_i acts freely on X. $Sh_K(G,X)_{\mathbb{C}}$ is a "fine moduli space"

$K_2 \triangleleft K_1$, normal subgp.

$$\begin{array}{c} Sh_{K_2}(G,X)_{\mathbb{C}} \\ \downarrow \\ Sh_{K_1}(G,X)_{\mathbb{C}} \end{array}$$

finite étale K_1/K_2.

$\varprojlim_K Sh_K(G,X)_{\mathbb{C}}$ $\leftarrow$ scheme qc sep.

$=: Sh(G,X)_{\mathbb{C}}$

$Sh(G,X)(\mathbb{C}) = G(\mathbb{Q}) \backslash X \times G(\mathbb{A}_f)$

In general (i.e. w/o assuming SVS).

$$G(\mathbb{Q})\backslash X \times G(\mathbb{A}_f)/K = G(\mathbb{Q})/Z(\mathbb{Q})\cap K \backslash X \times G(\mathbb{A}_f)/K$$

$$\varprojlim G(\mathbb{Q})\backslash X \times G(\mathbb{A}_f)/K = \frac{G(\mathbb{Q})}{Z(\mathbb{Q})}\backslash X \times \left(G(\mathbb{A}_f)/\overline{Z(\mathbb{Q})}\right)$$

Ex. $G = GL_2$, $X = \mathfrak{h}^{\pm}$,

$$K = \Gamma(N) = \{g \in GL_2(\hat{\mathbb{Z}}) : g \equiv 1 \mod N\}$$

$$Sh_{\Gamma(N)}(GL_2, \mathfrak{h}^{\pm})_{\mathbb{C}} = Y(N)_{\mathbb{C}}$$

when $N \geq 3$, $Sh_{\Gamma(N)}(GL_2, \mathfrak{h}^{\pm})_{\mathbb{C}}$ is a "fine moduli" space.

$N = 2$. $Y(2) \simeq \mathbb{P}^1 \setminus \{0, 1, \infty\}$.

The stack version of $Sh_{\Gamma(N)}(GL_2, \mathfrak{h}^{\pm})_{\mathbb{C}}$ is $Y(2) \times \mathbb{B}\mu_2$.

$$Y(2) \times \mathbb{B}\mu_2 = Y(4)/\left(\Gamma(2)/\Gamma(4)\right)$$

$$Y(1) \simeq \mathbb{P}^1 \setminus \{\infty\}.$$

$$\Gamma_0(N) = \left\{ g \in GL_2(\hat{\mathbb{Z}}) : g \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \mod N \right\}$$

$$\Gamma_0(N) \cap Z(\mathbb{Q}) \ni \begin{pmatrix} \pm 1 & \\ & \pm 1 \end{pmatrix}$$

$Sh_{\Gamma_0(N)}(GL_2, \mathfrak{h}^{\pm}) = Y_0(N)$ modular curve of level $\Gamma_0(N)$, is always a coarse moduli. Stacky version is a μ_2-gerb over $Y_0(N)$ when $N \geq 5$.

Let G be a linear alg. gp over no. field E. $\left(G(E) \subset G(\mathbb{A}_E) \text{ discrete} \right)$

Let S be a finite set of places of E, $\mathbb{A}_{E,S} = \prod_{v \in S} E_v$

$$\mathbb{A}_E^S = \prod_{v \notin S}{}' E_v$$

We say G satisfies weak (resp. strong) approximation w.r.t. S if $G(E)$ is dense in $G(\mathbb{A}_{E,S})$ (resp. $G(\mathbb{A}_E^S)$). (WA / SA)

<u>Thm</u> Let G be a linear alg gp $/E$

(1) If G is E-rat'l (e.g. G split reductive), then G satisfies weak approximation for every S.

(2) If G is semisimple, and is either simply-conn'd or adjoint type, then G satisfies (WA) for every S.

(3) In general, $\exists$ finite set S_0 of <u>finite</u> places of E s.t. G satisfies (WA) for every $S \cap S_0 = \phi$ (In particular, $G(E)$ is dense in $G(E \otimes \mathbb{R})$)

(4) G satisfies (SA) for S iff G is semisimple, simply-conn'd, & for every $\mathbb{Q}$-factor G_i of $\mathrm{Res}_{E/\mathbb{Q}} G$, $G_i(\mathbb{A}_{E,S})$ is non-compact.

Now let $X^+ \subset X$ be a conn'd component.

$$\begin{tikzcd}
 & G_{ad}(\mathbb{R}) \arrow[r, phantom, "\curvearrowright"] & D_{G,\mu} \arrow[r, "1:1 \text{ or } 2:1"] & G_{ad}(\mathbb{R})/K \\
G(\mathbb{R}) \arrow[r] & G_{ad}(\mathbb{R})_\perp \arrow[u, phantom, "\cup"] \arrow[r, phantom, "\curvearrowright"] & X \arrow[u, phantom, "\cup"] & \\
G(\mathbb{R})_+ \arrow[u] \arrow[r] & G_{ad}(\mathbb{R})^+ \arrow[u] \arrow[r, phantom, "\curvearrowright"] & X^+ \arrow[u, phantom, "\cup"] & \\
 & \text{conn'd component} & &
\end{tikzcd}$$

Let $G(\mathbb{Q})_+ = G(\mathbb{Q}) \cap G(\mathbb{R})_+$

$$G(\mathbb{Q})\backslash G(\mathbb{R}) \xleftarrow{\sim} G(\mathbb{Q})_+\backslash G(\mathbb{R})_+ \qquad \text{real approximation}$$

$$G(\mathbb{Q})\backslash X \times G(\mathbb{A}_f)/K \xleftarrow{\sim} G(\mathbb{Q})_+\backslash X^+ \times G(\mathbb{A}_f)/K$$

$$\pi_0\left(Sh_K(G,X)_{\mathbb{C}}\right) = G(\mathbb{Q})_+\backslash G(\mathbb{A}_f)/K$$

<u>Prop</u> If (G,X) satisfies (SV3), then $G(\mathbb{Q})_+\backslash G(\mathbb{A}_f)/K$ has a nat'l str. as a finite abelian gp.

In addition, if $G_{der} = G_{sc}$, then

$$G(\mathbb{Q})_+\backslash G(\mathbb{A}_f)/K \xrightarrow{\sim} G_{ab}(\mathbb{Q})^+\backslash G_{ab}(\mathbb{A}_f)/K_{ab}$$

where $G_{ab} = G/G_{der}$, $G_{ab}(\mathbb{Q})^+ = G_{ab}(\mathbb{Q}) \cap G_{ab}(\mathbb{R})^+$

$K_{ab} = \mathrm{Im}(K \to G_{ab}(\mathbb{A}_f))$ $\qquad G_{ab}(\mathbb{R})^+ = \mathrm{Im}\left(Z_G(\mathbb{R}) \to G_{ab}(\mathbb{R})\right)$

<u>Sketch of proof</u>.

(SA) wrt. $S = \{\infty\}$

$$\begin{tikzcd}
G_{sc} \arrow[r] \arrow[rr, bend right, "\pi"'] & G_{der} \arrow[r] & G \arrow[r] & G_{ab} \\
T_{sc} \arrow[u] \arrow[r] & T_{der} \arrow[r] & T \arrow[r] \arrow[u, phantom, "\cup"] & T_{ab} \arrow[u, phantom, "\shortparallel"]
\end{tikzcd}$$

$$G(\mathbb{Q})\backslash G_{sc}(\mathbb{A}_f)/K$$

$G(\mathbb{A}_f)/\pi\left(G_{sc}(\mathbb{A}_f)\right)$ is an abelian group

$G/G_{sc} \simeq T/T_{sc}$ as stacks

SA: $G_{sc}(\mathbb{A}_f) = G_{sc}(\mathbb{Q}) \cdot K$.

Lecture 15 (G, X) Shimura datum

$$\begin{tikzcd}
X^+ \arrow[r, phantom, "\subset"] & X \arrow[r, phantom, "\subset"] & D_{G,\mu} \arrow[r, phantom, "\subset"] & D_{G_{ad},\mu_{ad}} \arrow[r] & G_{ad}(\mathbb{R})/K \\
G_{ad}(\mathbb{R})^+ \arrow[u, two heads] \arrow[r] & G_{ad}(\mathbb{R})_1 \arrow[u] \arrow[rr, hook] & & G_{ad}(\mathbb{R}) \arrow[u, phantom, "\circlearrowleft \text{ transitively}"] & \\
G(\mathbb{R})^+ \arrow[u, two heads] \arrow[r, hook] & G(\mathbb{R})_+ \arrow[u] \arrow[r] & G(\mathbb{R}) \arrow[u] & &
\end{tikzcd}$$

$$\mathrm{Stab}_{G_{ad}(\mathbb{R})_1}(\pi_0(X)) = G_{ad}(\mathbb{R})^+$$

$$\Uparrow$$

$x \in D_{G_{ad},\mu_{ad}}$, $\mathrm{Stab}_{G_{ad}(\mathbb{R})}(x) = \{ghg^{-1} = h\}$ is conn'd

<u>Prop.</u> $\pi_0(Sh_K(G,X)_{\mathbb{C}}) \simeq G(\mathbb{Q})_+ \backslash G(\mathbb{A}_f)/K$ finite

- When (SV3) holds, this set has a natural abelian gp str.
- In addition, if $G_{der} = G_{sc}$, then $G(\mathbb{Q})_+ \backslash G(\mathbb{A}_f)/K \simeq G(\mathbb{Q})^+ \backslash G_{ab}(\mathbb{A}_f)/\nu(K)$

where $G_{ab} = G/G_{der}$, $\nu(K) = \mathrm{Im}(K \to G_{ab}(\mathbb{A}_f))$

$G_{ab}(\mathbb{Q})^+ = G_{ab}(\mathbb{Q}) \cap G_{ab}(\mathbb{R})^+$, $G_{ab}(\mathbb{R})^+ = \mathrm{Im}(Z_G(\mathbb{R}) \to G_{ab}(\mathbb{R}))$

$G_{sc}(\mathbb{Q}) \to G(\mathbb{Q})_+ = G(\mathbb{Q}) \cap G(\mathbb{R})_+$.

$$G(\mathbb{R})/G(\mathbb{R})_+ \xleftarrow[\sim]{\text{real approx.}} G(\mathbb{Q})/G(\mathbb{Q})_+$$

<u>Fact</u> $\pi_0 G_{sc}(\mathbb{R}) = \{1\}$

$\Uparrow$

finite $\pi_0 G(\mathbb{R})$

$\rightsquigarrow G_{ad}(\mathbb{R})_1/G_{ad}(\mathbb{R})^+ \simeq G_{ad}(\mathbb{Q})_1/G_{ad}(\mathbb{Q})^+$

$$\begin{tikzcd} G_{sc} \arrow[r, "\pi"] & G \\ T_{sc} \arrow[u, hook] \arrow[r] & T \arrow[u] \end{tikzcd}$$

$G/G_{sc} \xleftarrow{\sim} T/T_{sc}$. as stacks

(If $G_{sc} = G_{der}$, $G/G_{sc} = G_{ab}$)

(2) $G_{sc}(\mathbb{A}_f) = G_{sc}(\mathbb{Q})K'$, $\forall K'$ open cpt (SV3)

$\pi G_{sc}(\mathbb{A}_f) \subset G(\mathbb{Q})_+ K$

(3) $G_{sc} = G_{der}$, $\Rightarrow$ $H^1(\mathbb{Q}_p, G_{sc}) = 1 \overset{G_{der}=G_{sc}}{\Rightarrow} G(\mathbb{Q}_p)/G_{sc}(\mathbb{Q}_p) \simeq G_{ab}(\mathbb{Q}_p)$

$G(\mathbb{A}_f)/\pi(G_{sc}(\mathbb{A}_f)) = G_{ab}(\mathbb{A}_f)$

Ex. $G = GSp_{2g}$, $X = \mathfrak{h}_g^{\pm} = \{ Z \in M_g(\mathbb{C}) : Z = Z^t, \ \mathrm{Im}(Z) > 0 \text{ or } < 0 \}$

$G_{der} = Sp_{2g}$

$Z = \mathbb{G}_m$, $G_{ab} = \mathbb{G}_m$ $K = \Gamma(N) = \{ A : A \equiv 1 \bmod N \} \subset G(\hat{\mathbb{Z}})$

$$\begin{tikzcd} Z \arrow[r] \arrow[d, phantom, "\wr"] & G_{ab} \arrow[d, phantom, "\wr"] \\ \mathbb{G}_m \arrow[r] & \mathbb{G}_m \end{tikzcd}$$

$z \mapsto z^2$

$\nu\Gamma(N) = 1 + N\hat{\mathbb{Z}}$

$G_{ab}(\mathbb{Q})^+ = \mathbb{Q}^{>0}$

$$\pi_0\left(Sh_{\Gamma(N)}(GSp_{2g}, \mathfrak{h}_g^{\pm})\right) = \mathbb{Q}^{>0} \backslash \mathbb{A}_f^{\times}/(1 + N\hat{\mathbb{Z}})$$

$\Gamma_1(N)$

$\simeq (\mathbb{Z}/N\mathbb{Z})^{\times}$

$g = 1$, $\pi_0(Y(N)) = (\mathbb{Z}/N\mathbb{Z})^{\times}$

$\| $

$\pi_0(Y_1(N)) =$

$\pi_0(Y_0(N)) = 1$.

$Sh_K(GSp_{2g}, \mathfrak{h}_g^{\pm})_{\mathbb{C}}$

For S qcqs scheme, $\mathbb{P} \overset{\text{subring}}{\subset} \mathbb{Q}$

- $Av_{g,\mathbb{P}}(S)$ = cat. of abelian schemes $/S$ of rel. dim g, up to $\mathbb{P}$-isog.

  obj: $(A \to S)$

  mor: $\mathrm{Hom}_{Av_{g,\mathbb{P}}(S)}(A, B) = \mathrm{Hom}(A, B) \otimes \mathbb{P}$

  isom. in $Av_{g,\mathbb{P}}(S)$ are called homom. up to $\mathbb{P}^\times$-isogeny

- $A \rightsquigarrow A^\vee$

- A polarization is $(\lambda : A \to A^\vee) \in \mathrm{Hom}(A, A^\vee) \otimes \mathbb{P}$ s.t. for some $n \in \mathbb{Z}$, $n\lambda$ is étale locally on S induced by an ample line bundle.

A $\mathbb{P}$-principal polarisation λ is a polarization which is also an isom in $Av_{g,\mathbb{P}}(S)$.

$Av^{pp}_{g,\mathbb{P}}(S)$ p.p.a.v. up to $\mathbb{P}$-isog.

obj: (A, λ_A) $A \in Av_{g,\mathbb{P}}(S)$, λ_A p.p.

$$\downarrow$$

$$(B, \lambda_B)$$

morphisms $(A, \lambda_A) \to (B, \lambda_B)$ are $A \xrightarrow{f} B$, f is an isom in $Av_{g,\mathbb{P}}(S)$

s.t. $\lambda_B^{-1} \circ (f^\vee)^{-1} \circ \lambda_A = ct$ for some (Zariski) locally constant function $c : S \to \mathbb{P}^\times$.

$$S \longmapsto Av^{pp}_{g,\mathbb{P}}(S) \quad \text{groupoid}$$

$$\boxed{\begin{array}{l} Av^{pp}_{g,\mathbb{Z}}(S) \\ Av^{pp}_{g,\mathbb{Q}}(S) \\ Av^{pp}_{g,\mathbb{Z}_{(p)}}(S) \end{array}}$$

<u>Thm</u>. $Sh_{\Gamma(1)}(GSp_{2g}, h_g^{\pm})(\mathbb{C}) \xrightarrow{\sim} Av^{pp}_{g,\mathbb{Z}}(\mathbb{C})$

$$Sh_{\Gamma(1)}(GSp_{2g}, h_g^{\pm 1})_{\mathbb{C}} \xrightarrow{\sim} Av^{pp}_{g,\mathbb{Z}} \otimes \mathbb{C}$$

$N \geq 3$.

$$Sh_{\Gamma(N)}(GSp_{2g}, h_g^{\pm})(\mathbb{C})$$

$$\simeq \left\{ (A, \lambda, \eta) : \begin{array}{l} (A, \lambda) \in Av^{pp}_{g,\mathbb{Z}}(\mathbb{C}) \\ \eta = \eta_1 \oplus \eta_2 : (\mathbb{Z}/N)^{2g} \oplus \mathbb{Z}/N \xrightarrow{\sim} A[N] \oplus \mu_N \\ \text{s.t. } \lambda(\eta_1(x), \eta_1(y)) = \eta_2(\psi(x,y)) \end{array} \right\}$$

$A[N] \times A[N] \to \mu_N$

$$\simeq \left\{ (A, \lambda, \eta) : \begin{array}{l} (A, \lambda) \in Av^{pp}_{g,\mathbb{Q}}(\mathbb{C}), \\ \lambda : \Gamma(N)\text{-orbit of} \\ \eta : \mathrm{Isom}\left((\mathbb{A}_f^{2g} \oplus \mathbb{A}_f, \psi), (H_1(A, \mathbb{Q}) \otimes \mathbb{A}_f, \mathbb{Q}(1) \otimes \mathbb{A}_f, \lambda)\right) \end{array} \right\}$$

$$\simeq Sh_{\Gamma(N)}(GSp_{2g}, h_g^{\pm})(\mathbb{C})$$

G-torsor $\mathbb{Q}$, $(E, h : \mathbb{S} \to \mathrm{Aut}(E)_{\mathbb{R}}, \eta : E_{\mathbb{A}_f} \xrightarrow{\sim} G_{\mathbb{A}_f}$ K-orbit$)$

$$Sh_{\Gamma(N)}(GSp_{2g}, \mathfrak{h}_g^{\pm})_{\mathbb{C}} \xrightarrow{\sim} (\quad).$$

S smooth, (A, λ, η), $(A, \lambda) \in Av_{g,\mathbb{Q}}^{pp}$

$$(H_1(A, \mathbb{Q}), H_{dR}(A/S), F, \overset{\eta \cdots}{\cdots})$$

$$\begin{tikzcd}
 & S^{an} & \\
\tilde{S}^{an} \arrow[r] \arrow[d] & \mathfrak{h}^{\pm 1} \arrow[d] & \\
S^{an} \arrow[r] & Sh_{\Gamma(N)}(GSp_{2g}, \mathfrak{h}^{\pm 1})^{an}_{\mathbb{C}} &
\end{tikzcd}$$

$$S \longrightarrow Sh_{\Gamma(N)}(GSp_{2g}, \mathfrak{h}^{\pm 1})_{\mathbb{C}}$$

$$Av_{g,\mathbb{Z}}^{pp} \longrightarrow Sh_{\Gamma(N)}(GSp_{2g}, \mathfrak{h}^{\pm})$$

<u>Lecture 16</u>. $\mathbb{Z} \subset \mathbb{P} \subset \mathbb{Q}$, $\hat{\mathbb{Z}}^{\mathbb{P}} = \prod_{p \in \mathbb{P}^\times} \mathbb{Z}_p$, $\mathbb{A}_f^{\mathbb{P}} = \hat{\mathbb{Z}}^{\mathbb{P}} \otimes \mathbb{Q}$

S qcqs, $Av_{g,\mathbb{P}}^{pp}(S)$ principally polarized ab. var. $/S$ up to $\mathbb{P}$-isogeny

$$\| $$

$$\{(A, \underset{A \to A^\vee}{\lambda}) : \cdots\}$$

S qcqs, $Loc_{\mathbb{A}_f^{\mathbb{P}}}(S) = \{ \mathcal{E} : \mathbb{A}_f^{\mathbb{P}}\text{-local systems on } S_{\acute{e}t} \}$

$$\Psi \qquad = \{ \mathbb{L} : \underset{\mathrm{Hom}(-,-) \otimes \mathbb{P}}{\hat{\mathbb{Z}}^{\mathbb{P}}\text{-local system}} \}$$

$\mathbb{A}_f^{\mathbb{P}}(1)$

S smooth over a field k.

$\mathrm{Loc}_{dR}^{\mathrm{Fil}}(S) = \{ (\mathcal{E}, \nabla, \mathrm{Fil}) ; (\mathcal{E}, \nabla)$ flat conn., Fil decreasing filtration of $\mathcal{E}$ satisfying Griffith trans. $\}$

S smooth $/\mathbb{C}$.

$\mathrm{Loc}_{B, \mathbb{P}}(S) = \{ \mathbb{P}$- Betti local system on $S^{an} \}$

S $\mathbb{P}$-scheme

$$\begin{tikzcd}
\mathrm{Av}_{g,\mathbb{P}}^{pp}(S) \arrow[r] \arrow[dr] \arrow[ddr, bend right] & \mathrm{Loc}_{\mathbb{A}_f^{\mathbb{P}}}(S) \\
 & \mathrm{Loc}_{dR}^{\mathrm{Fil}}(S) \\
 & \mathrm{Loc}_{B,\mathbb{P}}(S)
\end{tikzcd}$$

$(A, \lambda) \longmapsto \left(\prod_{p \in \mathbb{P}^\times} T_p(A) \right) \otimes \mathbb{P} =: V(A)^{\mathbb{P}}$

(A, λ)

$K^{\mathbb{P}}(GSp_{2g}(\mathbb{A}_f^{\mathbb{P}}))$

$$\begin{array}{c}
\mathrm{GIsom}(V \otimes \mathbb{A}_f^{\mathbb{P}}, V(A)^{\mathbb{P}}) / K^{\mathbb{P}} \\
\shortparallel \\
\{ f : V \otimes \mathbb{A}_f^{\mathbb{P}} \oplus \mathbb{A}_f^{\mathbb{P}} \xrightarrow{\sim} V(A)^{\mathbb{P}} \oplus \mathbb{A}_f^{\mathbb{P}}(1) \} \\
\psi \quad \longleftrightarrow \quad \lambda
\end{array}$$

e.g. $K^{\mathbb{P}} = GSp_{2g}(\hat{\mathbb{Z}}^{\mathbb{P}})$, $\Gamma(S, \mathrm{GIsom}(-,-)/K^{\mathbb{P}}) \simeq \hat{\mathbb{Z}}^{\mathbb{P}}$- lattices in $V(A)^{\mathbb{P}}$ self dual w.r.t. λ up to scalar

$S \mapsto \mathcal{A}_{g,\mathbb{Q},K}(S) = \{ (A,\lambda) \in \mathrm{Av}^{pp}_{g,\mathbb{Q}}(S),\ \eta \in \Gamma(S, \mathrm{GIsom}(-,-)/K) \}$

$K = \ker(\mathrm{GSp}_{2g}(\hat{\mathbb{Z}}) \longrightarrow \mathrm{GSp}_{2g}(\mathbb{Z}/N)) = \Gamma(N)$

$S \longmapsto \{ (A,\lambda) \in \mathrm{Av}^{pp}_{p,\mathbb{Z}}(S),\ \eta : (\mathbb{Z}/N)^{2g} \oplus \mathbb{Z}/N \xrightarrow{\sim} A[N] \oplus \mu_N \}$

$= \mathcal{A}_{g,n}(S)$ $\qquad \psi \longleftrightarrow \lambda$

<u>Th.</u> $\mathcal{A}_{g,\mathbb{Q},K} \otimes \mathbb{C} \simeq Sh_K(\mathrm{GSp}_{2g}, h^{\pm 1})$

$\mathcal{A}_{g,\mathbb{Q},\Gamma(N)} \simeq \mathcal{A}_{g,N} / \mathbb{Q}$

<u>Def</u> A Shimura datum (G,X) is called <u>of Hodge type</u> if $\exists$ embedding $G \hookrightarrow \mathrm{GSp}_{2g}$ sending $X \mapsto h^{\pm}$.

$\mu_X \longmapsto \mu_g$

<u>Lem.</u> Let $G \hookrightarrow \mathrm{GSp}_{2g}$ be a closed embedding sending a conj. class of minuscule cocharacter μ of G to μ_g. Then $\exists$ at most one Shimura datum (G,X) s.t. $\mu_X = \mu$ and $(G,X) \hookrightarrow (\mathrm{GSp}_{2g}, h^{\pm 1})$

---

finitely generated or not ?

$G \hookrightarrow \mathrm{GSp}_{2g}$, $\qquad \tau \subset \bigoplus_{r \geq 0} V^{\otimes 2r}(r)$

$G = \{ g \in \mathrm{GSp}_{2g} : gt = t,\ \forall t \in \tau \}$

<u>Q.</u> Is τ f.g. ?

<u>Rmk</u> To define G, can replace τ by a f.g. subalg.

$$\mathrm{Av}^{pp}_{g,\mathcal{O}} \ni (A,\lambda), \quad \eta : \tau \longrightarrow H_1^B(A^{an}/S^{an})^{\otimes}$$

$$\psi \longleftrightarrow \lambda$$

$$\eta \otimes \mathbb{C} : \tau \longrightarrow \mathrm{Fil}^0 H_1^{dR}(A_{\mathbb{C}}/S_{\mathbb{C}})$$

$$\mathrm{Isom}\left(\cancel{V_f \oplus \mathbb{A}_f}\ V(A) \oplus \mathbb{A}_f(1)\right) =: G\mathrm{Isom}\left((V,\eta_0),(V(A),\eta)\right).$$

$$\psi \longleftrightarrow \lambda$$

$$\eta_0 \otimes \mathbb{A}_f \longleftrightarrow \eta_{\mathbb{A}_f}$$

$$\downarrow$$

$$G\mathrm{Isom}\left((V,\eta_0),(V(A),\eta)\right)/K$$

smooth

$$\mathcal{A}_{g,\mathcal{O},\tau,K} : S/\mathbb{C} \longmapsto \Big((A,\lambda,\eta,\eta) : (A,\lambda) \in \mathrm{Av}^{pp}_{g,\mathcal{O}},\ \eta : \tau \longrightarrow H_1^B(A^{an}/S^{an})^{\otimes}$$

sending ψ to λ

$$\eta \in \Gamma(S, G\mathrm{Isom}(-,-)/K^p),$$

$$\left(H_1^{dR}(A_{\mathbb{C}}/S_{\mathbb{C}}),\ \eta\otimes\mathbb{C},\ \mathrm{Fil}^0\right)$$

G-split μ-filtration $\Big)$

<u>Th.</u>

$$\begin{array}{ccc} \mathcal{A}_{g,\mathcal{O},\tau,K} & \simeq & \mathrm{Sh}_K(G,X) \\ \downarrow & & \downarrow \\ \mathcal{A}_{g,\mathcal{O},K_{GSp}} \otimes \mathbb{C} & \simeq & \mathrm{Sh}_{K_{GSp}}(GSp_{2g}, h^{\pm 1}) \end{array}$$

Shimura var. of unitary type.

Let $E|F$ be a quadratic ext. of no. field, $* : E \longrightarrow E$.

V finite E-v.s., can also regarded as a f.d. $\mathbb{Q}$-v.s.,

$\iota : E \longrightarrow \mathrm{End}_{\mathbb{Q}}(V)$

<u>Lemma</u> $\exists$ canonical bijection

(SHF 1) a skew-sym $\mathbb{Q}$-bilinear form

$\psi : V \times V \longrightarrow \mathbb{Q}$ s.t. $\psi(\iota(a)v, w) = \psi(v, \iota(a^*)w)$

$\mathrm{Tr}_{E|\mathbb{Q}}$

(SHF 2) a $\mathbb{Q}$-bilinear $\phi : V \times V \longrightarrow E$ s.t. $\phi(av, w) = a\phi(v,w)$ $a \in E$

$\phi(w, v) = -\phi(v,w)^*$.

$$\{g \in GSp(V,\psi) : g\,\iota(a) = \iota(a)g\} \simeq \{g \in GL(V,E) : \phi(gv, gw) = c(g)\phi(v,w),\ c(g) \in \mathbb{Q}^\times\}$$

$$GU_{E/F}(V,\psi) \simeq GU_{E/F}(V,\phi)$$

$$\begin{array}{ccccc} G_{der} & \subset & G_1 & \subset & G \\ \| & & \| & & \\ \{g : c(g) = 1, \det(g) = 1\} & & \{g : c(g) = 1\} & & \\ \| & & \| & & \\ SU_{E/F}(V,\phi) & & U_{E/F}(V,\phi) & & \end{array}$$

<u>Lem</u>

Let (V,E) be as above. Choose $b \in E$, $b^* = -b$. Then we have a bijection

ϕ (SHF 2) skew-hermitian form ϕ

$\updownarrow$

(HF) hermitian form

$b\phi = \theta$

$\theta : V \times V \longrightarrow E$

$\theta(av, w) = a\theta(v,w)$, $\theta(w,v) = \theta(v,w)^*$

Th. Let $E|F$ be a nontrivial CM extn. Let (V, ϕ) be a nondeg. E-skew-hermitian space. We write $[E:\mathbb{Q}]=2d$, $[F:\mathbb{Q}]=d$. $\dim_E V = n$, $g = dn$. Let $G = GU(V,\phi) \hookrightarrow GSp(V, \underset{\mathrm{Tr}\phi}{\overset{\shortparallel}{\psi}})$

(1) $\exists!$ Shimura datum (G, X) s.t. ρ induces an embedding

$$(G, X) \hookrightarrow (GSp(V), h^{\pm 1})$$

For $h \in X$, giving a HS on V, let r_σ be the dim. of the σ-comp. of $(E \otimes \mathbb{C})$-mod $V_{\mathbb{C}}^{-1,0}$. Then $r_\sigma + r_{\sigma^c} = n$.

(2) $Sh_K(G, X)_{\mathbb{C}} = \{ (A, \lambda, \iota, \eta) : (A, \lambda) \in Av_{g,\mathbb{Q}}^{pp}$

$\iota : E \to \mathrm{End}\, A \otimes \mathbb{Q}$

- sending $* \to *$
- σ-comp. of $(E \otimes \mathbb{C})$-mod on $\mathrm{Lie}\, A$ has rank r_σ $\}$

$\eta \in \Gamma(S, G\mathrm{Isom}(-)/K)$

unitary SV.

$$\det(T - a \mid \mathrm{Lie}\, A) = \prod_\sigma (T - \sigma(a))^{r_\sigma}.$$

Lecture 17. So far, (G, X) Shimura datum, $K \subset G(\mathbb{A}_f)$ neat

$\rightsquigarrow Sh_K(G, X)_{\mathbb{C}}$ smooth quasi proj. var.

If $G = T$, $X = \{h : \mathbb{S} \to T_{\mathbb{R}}\}$ $\left(\begin{smallmatrix} SV1 \\ SV2 \end{smallmatrix}\right)$ ✓

$$Sh_K(T,\{h\})_{\mathbb{C}} = T(\mathbb{Q})\backslash T(\mathbb{A}_f)/K \qquad \text{finite set}$$

$$Sh(T,\{h\})_{\mathbb{C}} = \varprojlim_K Sh_K(T,\{h\}) = \overline{T(\mathbb{Q})}\backslash T(\mathbb{A}_f)$$

$$\overline{\mathbb{Q}} \subset \mathbb{C}$$

$E = E(G,X) \subset \overline{\mathbb{Q}}$ reflex field is the field of def'n of $\{\mu_x\}$

$$\mathrm{Hom}(\mathbb{G}_{m,\mathbb{C}}, G_{\mathbb{C}})/G_{\mathbb{C}} \simeq \mathrm{Hom}(\mathbb{G}_{m,\mathbb{C}}, T_{\mathbb{C}})/W \simeq \mathrm{Hom}(\mathbb{G}_m, \overline{\mathbb{Q}}, T_{\overline{\mathbb{Q}}})/W$$

$$\mathrm{Gal}(\overline{\mathbb{Q}}|E) = \{\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}|\mathbb{Q}) : \gamma\mu = w\mu \text{ for some } w \in W\}$$

$Sh(T,\{h\})_{\mathbb{C}}$ has a nat'l str. as a scheme $/E$, defined as

$$\begin{tikzcd}
\mathrm{Gal}(\overline{\mathbb{Q}}|E) \arrow[r, two heads] & \mathrm{Gal}(E^{ab,\infty}|E) & \overline{E^\times}\backslash \mathbb{A}_E^{\times,\infty} \arrow[l, "\mathrm{Art}", "\sim"'] \arrow[r] \arrow[d] & \overline{T(\mathbb{Q})}\backslash T(\mathbb{A}_f) \arrow[d] \\
 & & & T(\mathbb{Q})\backslash T(\mathbb{A}_f)/K
\end{tikzcd}$$

max. abelian ext'n of E split at all real places

$$\mu : \mathbb{G}_m \longrightarrow T_{\overline{\mathbb{Q}}} \text{ defined } /E \iff \mathrm{Res}_{E|\mathbb{Q}}\, \mathbb{G}_m \longrightarrow T$$

$$\begin{tikzcd}
\mathbb{A}_E^{\times,\infty} \arrow[r] \arrow[d] & T(\mathbb{A}_f) \arrow[d] \\
\overline{E^\times}\backslash \mathbb{A}_E^{\times,\infty} \arrow[r] & \overline{T(\mathbb{Q})}\backslash T(\mathbb{A}_f)
\end{tikzcd}$$

$\rightsquigarrow$ $Sh_K(T,\{h\})$ smooth E-scheme, $\quad Sh_K(T,\{h\}) \otimes_E \mathbb{C} = Sh_K(T,\{h\})_{\mathbb{C}}$

Th. (Canonical model) Let (G, X) be a Shimura datum ((SV1), (SV2)), $K \subset G(\mathbb{A}_f)$ neat. $\exists!$ smooth quasi-proj. var. $Sh_K(G, X)$ over $E := E(G, X) \subset \overline{\mathbb{Q}} \subset \mathbb{C}$ s.t.

(1) compatible w/ varying K

$$\rightsquigarrow \quad Sh(G, X) = \varprojlim_K Sh_K(G, X) \circlearrowleft G(\mathbb{A}_f)$$

(2) $$Sh(G, X) \otimes_E \mathbb{C} \simeq Sh(G, X)_{\mathbb{C}} \qquad G(\mathbb{A}_f)\text{-equiv.}$$

(3) for every $(T, \{h\}) \to (G, X)$ where T is a torus,

$$Sh(T, \{h\})_{\mathbb{C}} \longrightarrow Sh(G, X)_{\mathbb{C}}$$ arises as the base change of

$$Sh(T, \{h\}) \longrightarrow Sh(G, X) \otimes_{E(G,X)} E(T, \{h\})$$

Rmk. $(G_1, X_1) \hookrightarrow (G_2, X_2)$

$E(G_1, X_1) \supset E(G_2, X_2)$

$$Sh(G_1, X_2) \hookrightarrow Sh(G_2, X_2) \otimes_{E(G_2, X_2)} E(G_1, X_1)$$

Recall $(G, X) = (GSp, \mathfrak{h}^{\pm})$

$$\mathcal{A}_{g,K} \otimes \mathbb{C} \simeq Sh_K(GSp_{2g}, \mathfrak{h}^{\pm})_{\mathbb{C}}$$

$\mathcal{A}_{g,K}/\mathbb{Q}$ $\quad S \mapsto \{(A,\lambda,\eta) : (A,\lambda) \in \mathrm{Av}^{pp}_{g,\mathbb{Q}}(S),\ \eta\ K\text{-level}\}$

In fact, $\mathcal{A}_{g,K}$ is a smooth $\mathbb{Q}$-scheme.

<u>Th.</u> $\mathcal{A}_{g,(K)}$ is the canonical model of $Sh_{(K)}(GSp, h^{\pm 1})$

A g-dim'l abelian var. $/\mathbb{C}$, $\quad G_{MT}(H_1(A,\mathbb{Q}))$ abelian

$\Leftrightarrow$ $\mathrm{End}^\circ A$ is a CM algebra

$\qquad\qquad\qquad\qquad \parallel$

$\qquad\qquad\qquad\qquad \prod E_i, \quad E_i$ CM field.

$$G_{MT} \subset G\left(\prod_i U_{E_i/F_i}(1)\right)$$

<u>Th.</u> (Shimura - Taniyama formula) Let A be a (simple for simplicity) CM-abelian var. $\bar{\mathbb{Q}} \subset \mathbb{C}$ $\quad (E, \Phi)$

$$\begin{tikzcd}
H_1(A,\mathbb{Q}) \otimes \mathbb{A}_f \simeq V(A) & & \\
\gamma \in \mathrm{Gal}(\bar{\mathbb{Q}}|\mathbb{Q}) \arrow[u, bend left] \arrow[d, "\wr", "\alpha_\gamma"'] & \gamma A = A \otimes_{\bar{\mathbb{Q}}, \gamma} \bar{\mathbb{Q}} \arrow[d, "\wr", "\gamma_*"'] & \text{CM type } (E, \gamma\Phi) \\
H_1(\gamma A, \mathbb{Q}) \otimes \mathbb{A}_f \simeq V(\gamma A) & &
\end{tikzcd}$$

free $\mathbb{A}^\infty_E$-mod of rank 1

Pick $H_1(A,\mathbb{Q}) \xrightarrow{\eta}_{\sim} H_1(\gamma A, \mathbb{Q})$ as E-vector space. η is unique up to $E^\times$

well-defined

$$f(\gamma) \in E^\times \backslash (\mathbb{A}_E^\infty)^\times = \mathrm{Res}_{E|\mathbb{Q}}(\mathbb{Q}) \backslash \mathrm{Res}_{E|\mathbb{Q}}^{\mathbb{G}_m}(\mathbb{A}_f)$$

Suppose $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}} | E_\Phi)$ i.e. $\gamma \Phi = \Phi$. $\quad E_\Phi = E(\mu_\Phi)$

In this case, one can get η by an isogeny $A \dashrightarrow \gamma A$ $/\mathbb{C}$.

Then

$$\begin{tikzcd}
\mathrm{Gal}(\overline{\mathbb{Q}} | E_\Phi) \arrow[r, two heads] \arrow[ddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd] & \mathrm{Gal}(E_\Phi^{ab,\infty} | E_\Phi) \simeq E_\Phi^\times \backslash \mathbb{A}_{E_\Phi}^{\infty,\times} \arrow[r, two heads] & \mathbb{G}_{E_\Phi}(\mathbb{A}_f) \backslash \mathbb{G}_{E_\Phi}(\mathbb{Q}) \arrow[d] \\
 & & G_{MT}(A)(\mathbb{A}_f) \backslash G_{MT}(\mathbb{Q}) \arrow[d, hook] \\
 & & GU_F^{(1)}(\mathbb{A}_f) \backslash GU_F^{(1)}(\mathbb{Q}) \arrow[d] \\
 & & \mathrm{Res}_{E|\mathbb{Q}}^{\mathbb{G}_m}(\mathbb{A}_f) \backslash \mathrm{Res}_{E|\mathbb{Q}}(\mathbb{Q})
\end{tikzcd}$$

$f(\gamma)$ is the image of γ

under this sequence of maps

<u>Th</u> $(G, X) \hookrightarrow (GSp_{2g}, h^{\pm 1})$ $\quad \iota : \mathcal{T} \to V^\otimes$

$G = \{ g \in GSp : gt = t, t \in \mathcal{T} \}$

$$\mathrm{Sh}_K(G, X) : \mathrm{Sm}_E \longrightarrow \mathrm{Sets}$$

$$S \longmapsto \{ (A, \lambda, \iota, \eta) \}$$

- $(A, \lambda) \in \mathrm{Av}_{g,\mathbb{Q}}^{pp}$
- $\iota = (\iota_{dR}, \iota_{et})$: $\iota_{dR} : \mathcal{T} \longrightarrow H_1^{dR}(A/S)^\otimes$, $\iota_{dR}(\tau) \subset \mathrm{Fil}^0(-)$

$$\psi \longmapsto \lambda$$

$\iota_{et} : \mathcal{T} \longrightarrow V(A)^\otimes$

$-\eta \in \Gamma(S, \mathrm{GIsom}(-,-)/K)$

Satisfying (1) $(H_1^{dR}, \tau_{dR}, \mathrm{Fil})$ G-split μ-filtration

(2) $H_1^B(\sigma A, \mathbb{Q}) \otimes \mathcal{O}_S \xrightarrow{\sim} H_1^{dR}((A/S)_{\mathbb{C}})$ $\quad E \xhookrightarrow{\sigma} \mathbb{C}$

$\tau_{dR}(\tau)$ factors through $\tau_B : \tau \to H_1^B(\sigma A, \mathbb{Q})^{\otimes}$

and $\tau_B = \tau_{et}$ under Betti - étale comparison

<u>Rmk</u> Under every $\sigma' : E \hookrightarrow \mathbb{C}$, (2) will hold.

"Hodge cycles are absolute Hodge"

Fix a rep'n $G_{\mathbb{Q}_p} \hookrightarrow GL(V_{\mathbb{Q}_p})$, $K_p \subset G(\mathbb{Q}_p) \subset GL(V_{\mathbb{Q}_p})$

for some lattice L, $L \subset V_{\mathbb{Q}_p}$

$$\begin{array}{ccc} Sh_{K_p' K^p}(G, X) & & \\ K_p/K_p' \circlearrowright \downarrow & & \\ Sh_{K_p K^p}(G, X) & \rightsquigarrow & \mathbb{Z}/p^n\mathbb{Z}\text{-local system} \\ \downarrow & & \\ GL(L/p^n L) & & \end{array}$$

$K_p^{(n)} = \{ g \in K_p : g$ acts trivially on $L/p^n L \}$

$\rightsquigarrow$ étale $\mathbb{Z}_p$- local system on $Sh_K(G, X)$

First step to prove the existence of canonical model:

$Sh_K(G, X)_{\mathbb{C}} \otimes_{\mathbb{C}, \gamma} \mathbb{C}$ is a "Shimura variety".

Pasero